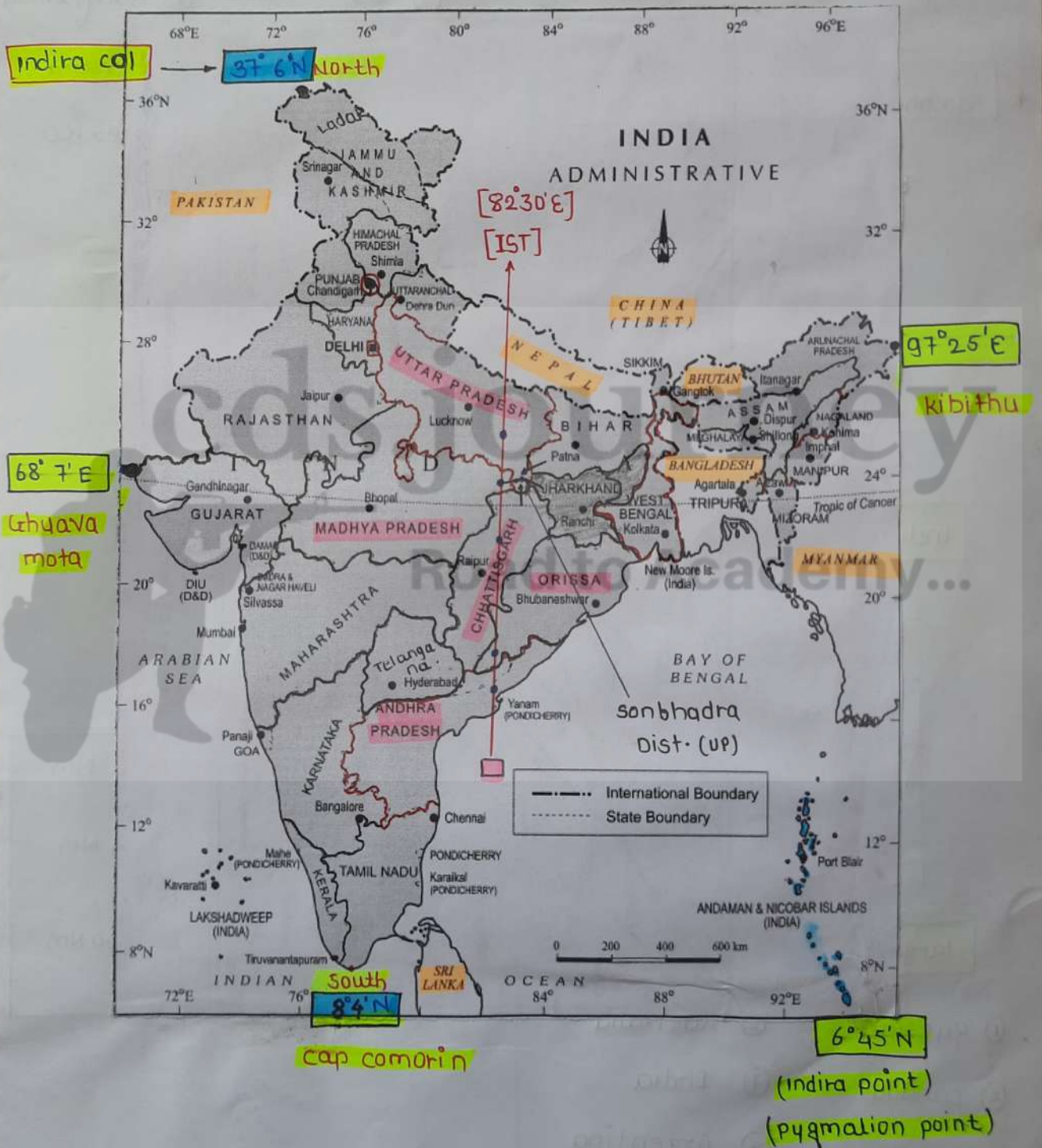


INDIAN GEOGRAPHY



* latitudinal and longitudinal extent
of India is about - 30 degree.

• North - south - 3214 km

• east - west - 2933 km

* sonbhadra district → UP



border share with

- mp
- chhattisgarh
- Jharkhand
- Bihar

Indian subcontinent

- India
- Pakistan
- Bhutan
- Bangladesh
- Nepal

largest countries by Area

- | | |
|----------|--------------|
| ① Russia | ⑥ Australia |
| ② Canada | ⑦ India |
| ③ USA | ⑧ Argentina |
| ④ China | ⑨ Kazakhstan |
| ⑤ Brazil | ⑩ Algeria |

countries in the world-popula-

- | | |
|-------------|--------------|
| ① India | ⑥ Nigeria |
| ② China | ⑦ Brazil |
| ③ US | ⑧ Bangladesh |
| ④ Indonesia | ⑨ Russia |
| ⑤ Pakistan | ⑩ Mexico |

International Boundaries

- 1) Bangladesh
- 2) China
- 3) Pakistan
- 4) Nepal
- 5) Myanmar
- 6) Bhutan
- 7) Afg.

1 Nm = 1.8 km

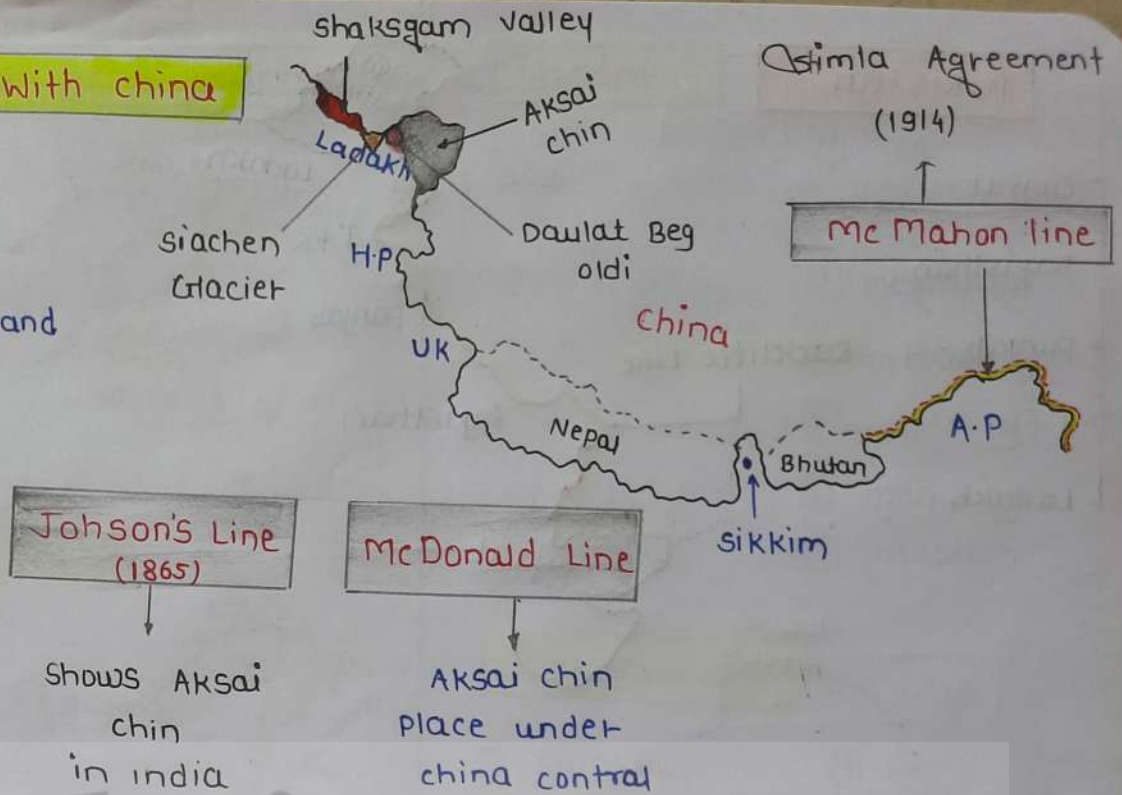
♦ territorial limits - 12 Nm

♦ contiguous zone - 24 Nm

♦ exclusive Economic - 200 Nm
Zone

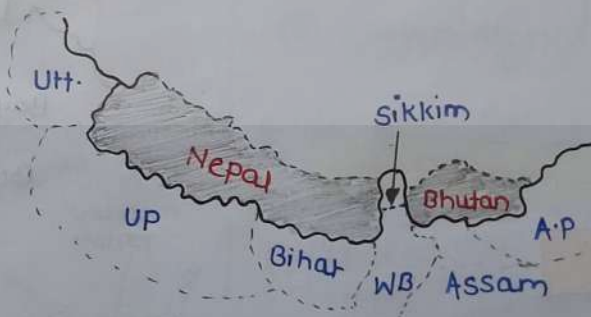
Border with china

- LAC
- Ladakh
 - H.P
 - Uttarakhand
 - Sikkim
 - A.P



Nepal

- Utt.
- UP
- Bihar
- W.B
- Sikkim



Bhutan

- ① Sikkim
- ② WB
- ③ A.P
- ④ Assam

Pakistan

- Gujarat

- Rajasthan

- Punjab

- J&K

- Ladakh

Raddcliffe Line

LOC

Ladakh

J&K

Punjab

Rajasthan

Gujarat

Bangladesh

- WB

- Assam

- Meghalaya

- Tripura

- Mizoram

WB

Assam

Meghalaya

Naga

Manipur

Mizoram

Bang.

Bay of Bengal

Myanmar

- Arunachal Pradesh

- Nagaland

- Manipur

- Mizoram

A.P.

Nagaland

Manipur

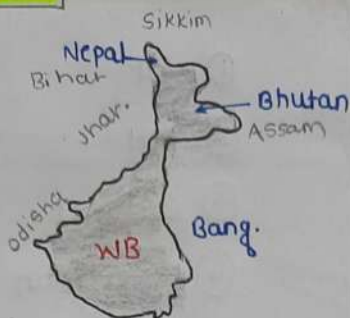
Mizoram

Bang.

Myanmar

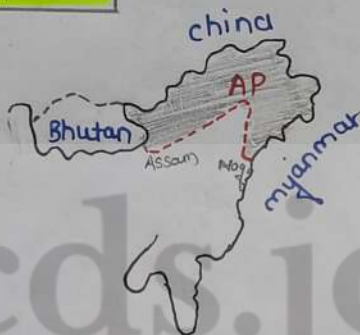
West Bengal

- Nepal
- Bhutan
- Bangladesh



Arunachal Pradesh

- China
- Bhutan
- Myanmar



Sikkim

- China
- Nepal
- Bhutan



Ladakh

- Pakistan
- China
- Afghanistan



Mizoram

- Bangladesh
- Myanmar

- State
- 1. Assam
 - 2. Tripura
 - 3. Manipur

Assam

- Bhutan
- Bangladesh

- 1. WB
- 2. AP
- 3. Meghalaya
- 4. Tripura
- 5. Nagaland
- 6. Manipur
- 7. Mizoram

Uttar Pradesh

- 1. Rajasthan

- 9. Delhi

- 2. Haryana
- 3. Himachal Pradesh
- 4. Uttarakhand
- 5. Bihar
- 6. M.P.
- 7. Jharkhand
- 8. Chhattisgarh

sonbhadra district

Chhattisgarh

- 1. U.P.
- 2. M.P.
- 3. Maharashtra
- 4. Jharkhand
- 5. Odisha
- 6. A.P.
- 7. Telangana

Jharkhand

- 1. Bihar
- 2. U.P.
- 3. Chhat.
- 4. W.B. ✓
- 5. Odisha

Haryana

- ① Punjab
 - ② H.P
 - ③ Rajasthan
 - ④ Uttarakhand
 - ⑤ U.P
- three side with
• Delhi
- } Yamuna River
define border

Telangana

- ① Maharashtra
 - ② Chhattisgarh
 - ③ A.P
 - ④ Karnataka
- formation
↓
B.N Srikishna
Committee

West Bengal

- ① Odisha
 - ② Jharkhand
 - ③ Bihar
 - ④ Sikkim
 - ⑤ Assam
- Kolkata
↓
• 7th largest city
• 3rd largest
metropolitan
area.

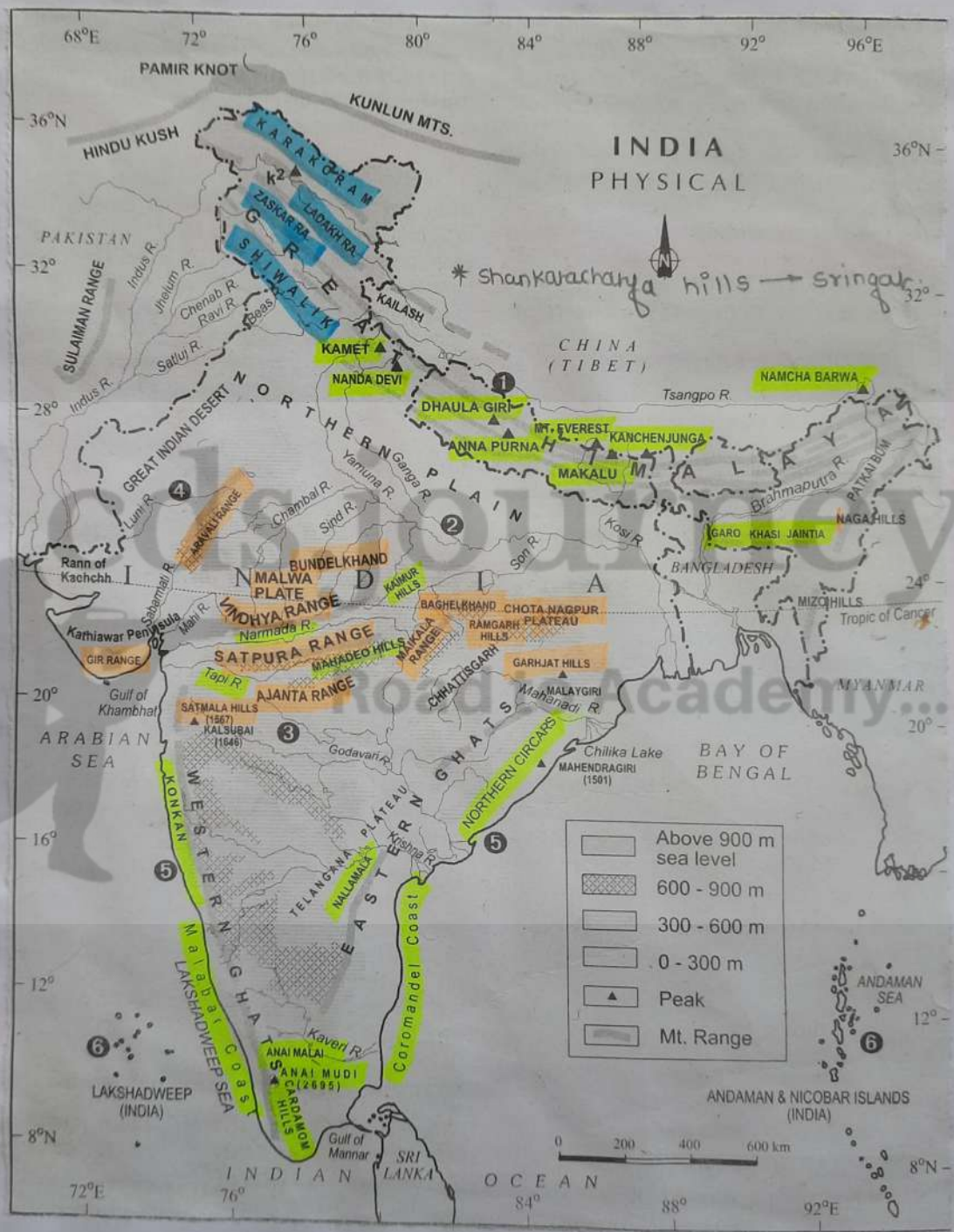
Maharashtra

- ① Gujarat
 - ② M.P
 - ③ Chha.
 - ④ Telangana
 - ⑤ Karnataka
 - ⑥ Goa
- Dadra & Nagar Haveli
+ Diu + Daman.

Madhya Pradesh

- ① UP
- ② MH
- ③ Gujarat
- ④ Rajasthan
- ⑤ Chhattisgarh

PHYSIOGRAPHY

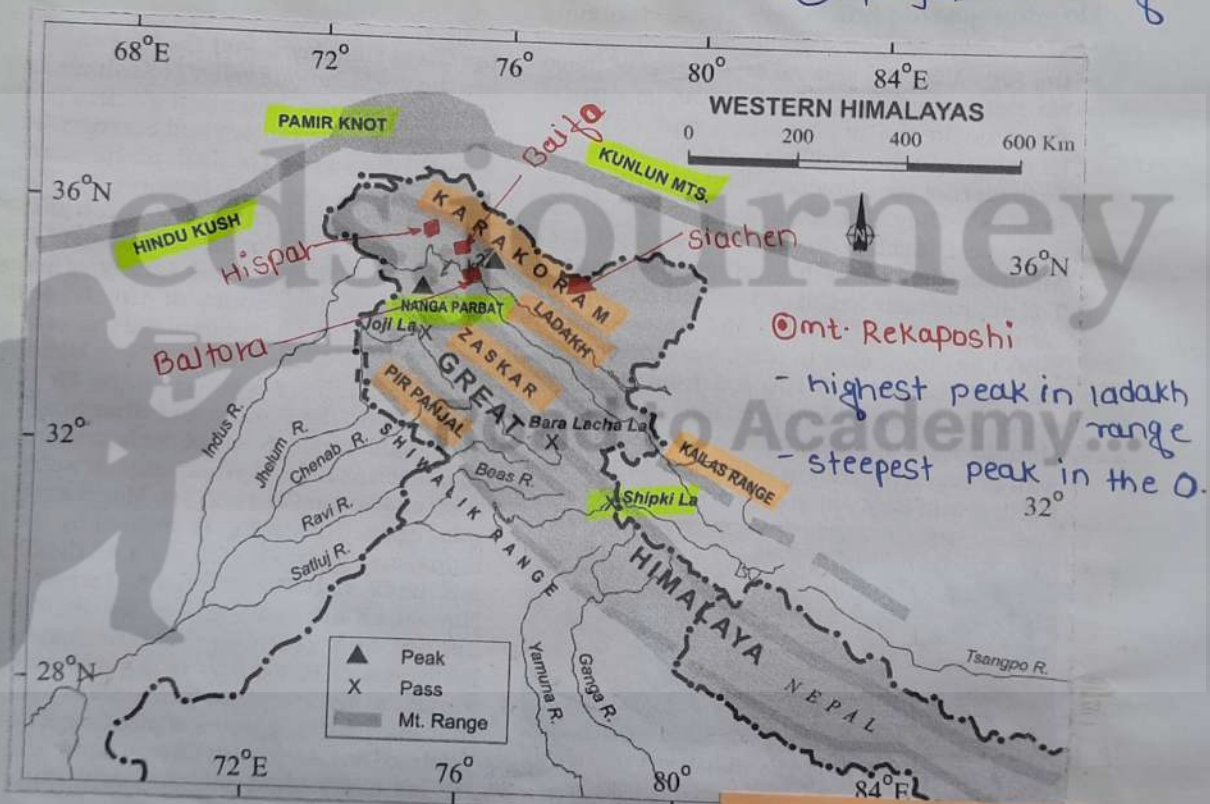


Himalayas

- ① Trans Himalaya
- ② Great Himalaya
- ③ mid / Lesser Himalaya
- ④ Shivalik
- ⑤ purvanchal Hills

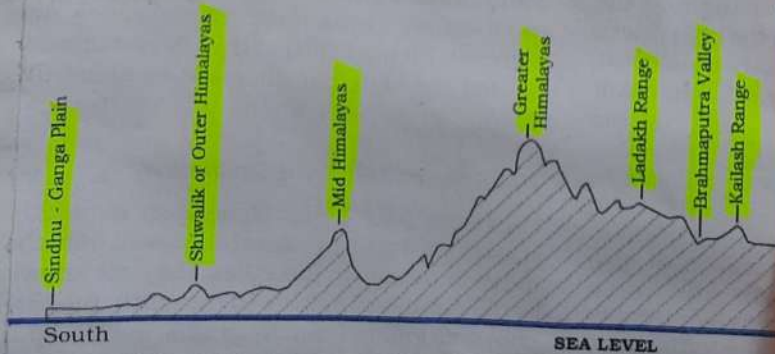
regional division

- ① kashmir Himalaya
- ② nepal Himalaya
- ③ Assam Himalaya
- ④ punjab Himalaya



● Mt. Rekaposhi
 - highest peak in Ladakh range
 - steepest peak in the O.

STRUCTURE AND PHYSIOGRAPHY



Trans Himalaya

- State/ UT
- ① Ladakh
 - ② J & K
 - ③ H.P

Avg. elevation - 3000m

mountain ranges

- Karakoram range
- Ladakh range
- Zaskar range
- Kailas range.



Region of world largest glaciers

- siachen
- Baltora
- Baltora
- Hisper

Kashmir Himalayas

Famous for karewa forma-

which are useful for cultivation of zafraan
local variety - saffron

karewas:

thick deposits of glacial clay & other materials embedded with moraines.

Great Himalaya

AP, Assam, Nepal
UK, H.P, J&K

about 2500 km extension

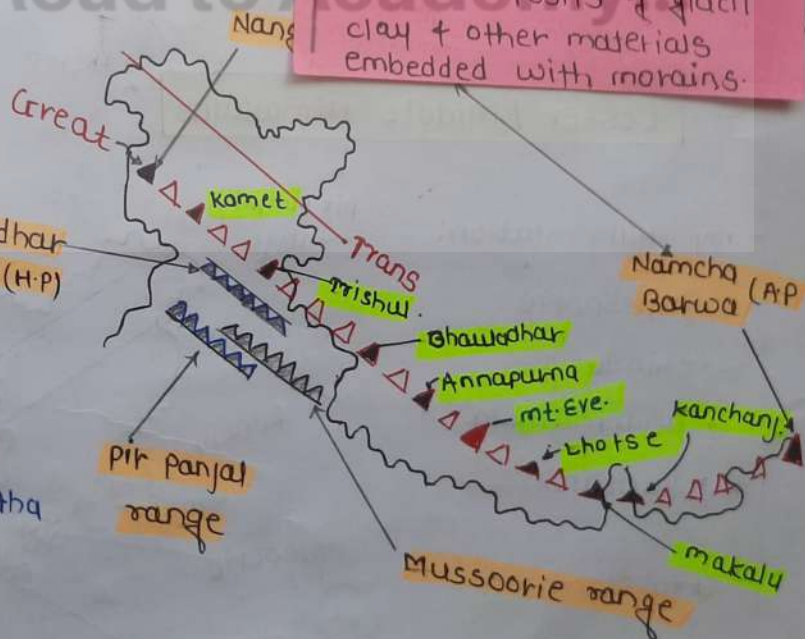
Nanga prabat to Namcha Barwa

highest peak - mt. Everest

Tibet
chomolungma.

Nepal
sagar matha

- Zoji La - Great Hima.
- Banihal - pir panjal
- Photu La - Zaskar
- Khardunga - Ladakh range.



* Great Himalayan range
mostly inhabited by
Bhotia's UK, SK, WB

Highest peak

- Utt - Nanda Devi
- WB - tiger hill
- Jhar - parasnath
- Gujrat - Sarkala
- MH - Kalsubai
- Nilgiri - Doddabetta
- Nag. - Saramati



• Lepcha
tribes
↓
Sikkim

Lesser / Middle Himalayas

J&K, H.P.,
UK

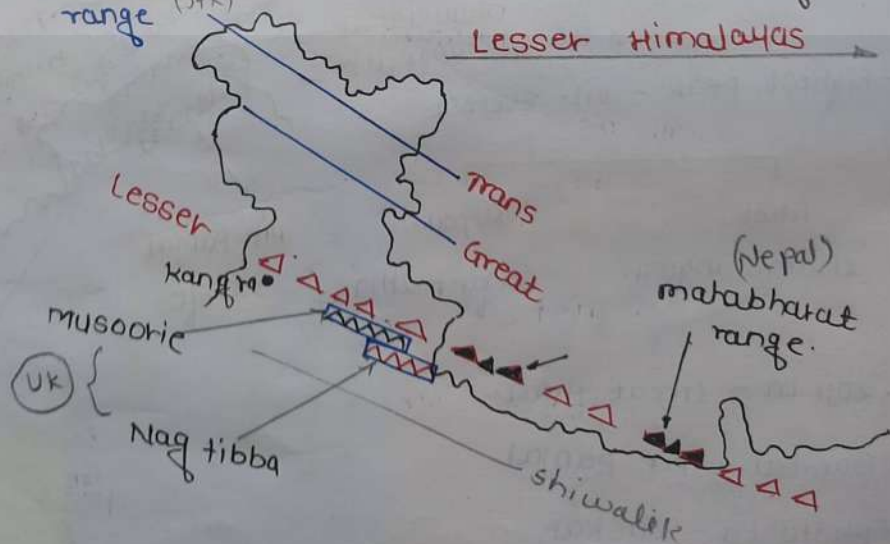
- imp. Hills station

- mussoorie
- Shimla
- Dharmashala
- Raosani

- Grassland

- Kashmir - marg
 - Gulmarg
 - Sonmarg
- UK - Bygatal - Gharwal
 - Panchdar - Kumaon.

pir panjal
range (J&K)



Great Himalayas

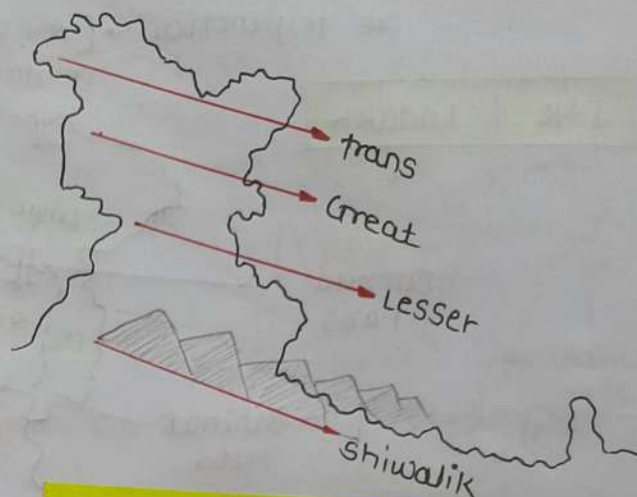
- Kashmir valley
- Kathmandu valley
- Kanga & Kulu valley

Lesser Himalayas

• coniferous
trees found.

Shiwalik / outer Himalaya

- thick forests
- rainfall (↓) east to west in shiwalik + Ganga plain.
- plain b/w lesser & shiwalik
↓
known as - Doons / Doon
 - Dheradoon - UK
 - Patlidoon - UP
 - Kotlidoon - Pok
- youngest part of the Himalayas



Some of the prominent ones from west to east are the

→ Monpa	→ Nishi
→ Daffla	→ Nagas
→ Abor	Most of this
→ Mishmi	communities practice
	Jhumming

Also known as shifting or slash and burn cultivation.

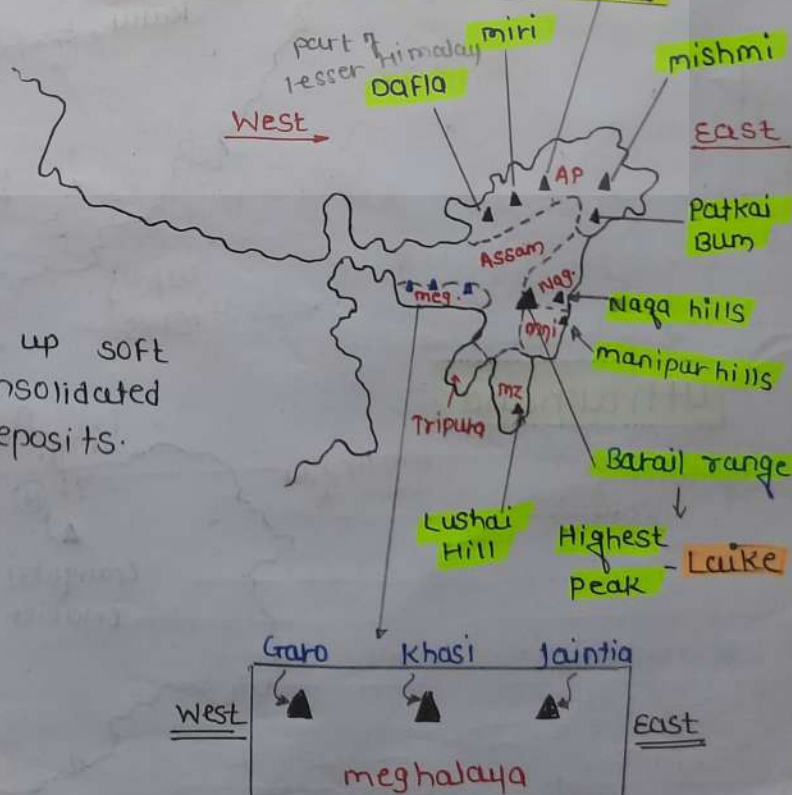
Purvanchal Hills

- highest peak
 - AP - Kangto
 - Manipur - Mt. Tempu
 - Mizoram - Blue Mountain
- Also known as molassis basin → made up soft unconsolidated deposits.

☆ highest peak state

- ▲ Kangto (Kanggardo) - A.P
- ▲ Mt. Saramati - Nag.
- ▲ Mt. Tempu (Mt. Iso) - Manip.
- ▲ Phawngpui (Blue Moun.) - Mizo.

• Muli Ghat Hills.



Passes

(Nubra valley)

trijunction → [A + C + I]
mintaka

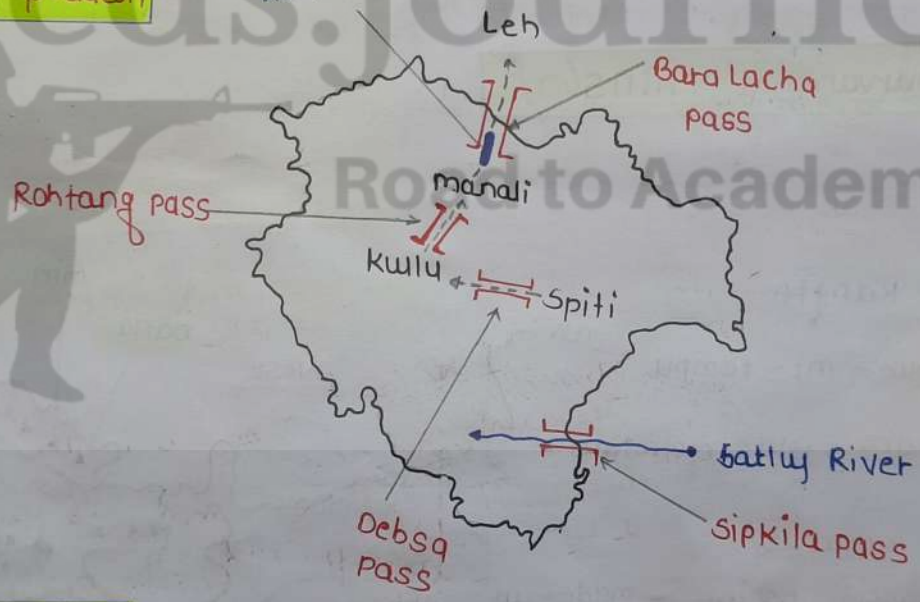
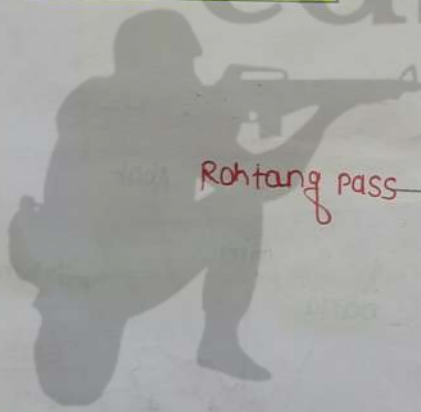
through - siachen glacier

J&K + Ladakh

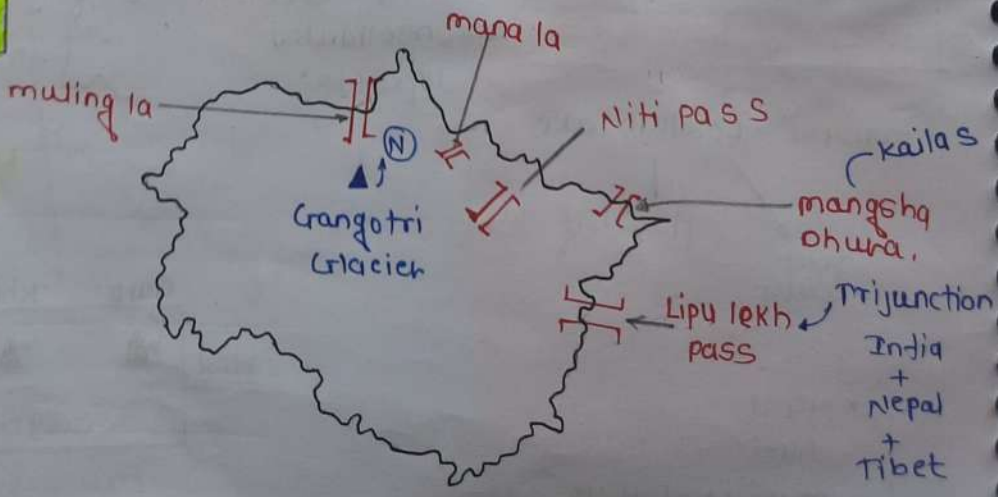


Himachal Pradesh

(manali to Leh)
Atal tunnel



Uttarakhand

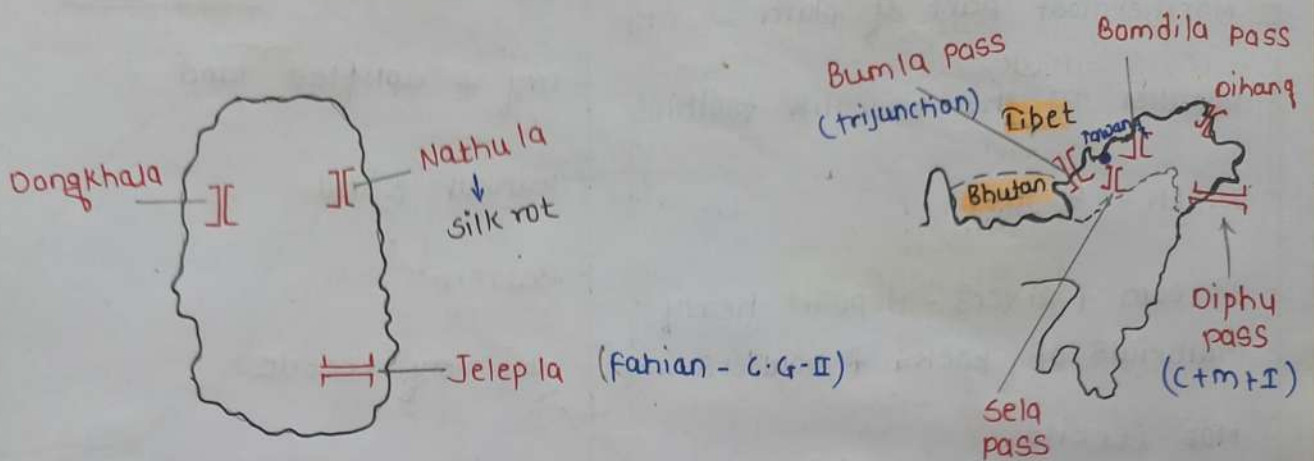


India + Nepal + Tibet

Sikkim

[A-371F]

Arunachal p.



Northern plains

- formed - Alluvial deposits.
- brought by
 - Indus
 - Ganga
 - Brahmap.

- extend - 3200km
- width - BLW 150-300km

divided into three major part

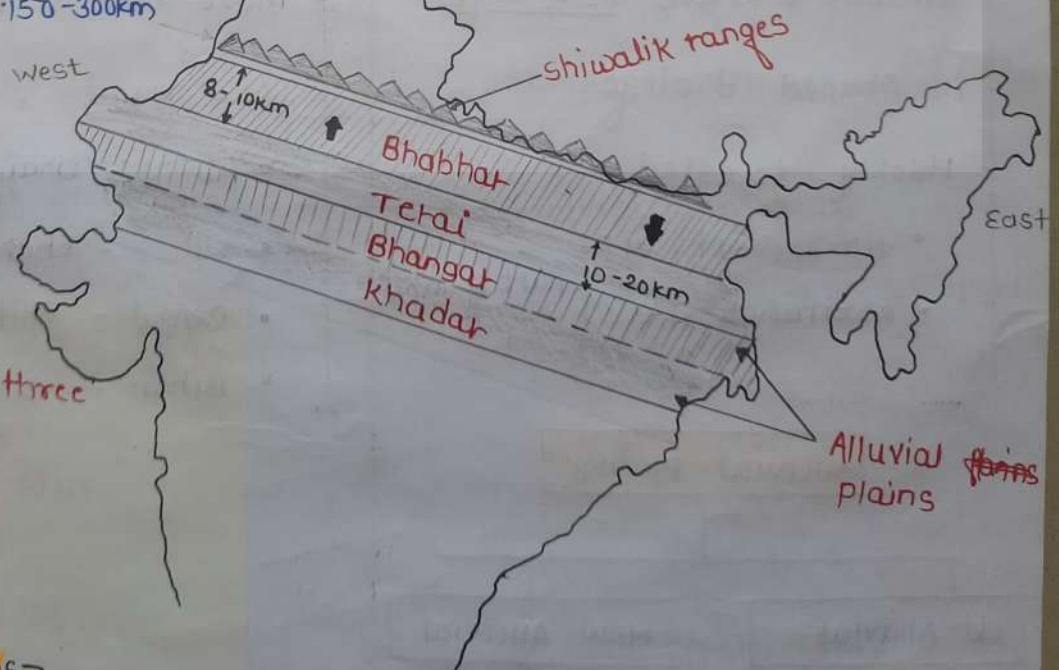
Bhabhar

Terai

Alluvial Plains

• Bhangar

• Khadar



Bhabar

- Northernmost part of plain
- parallel to the Shiwalik foothills
- Width - 8-10 km
- Stream & Rivers - deposit heavy materials of rocks & boulders
- Not Fertile
- Rivers - streams disappear.

Tarai

- South of Bhabar
 - Width - 10-20 km
 - Streams & Rivers re-emerge
 - ill shaped drainage
 - thickly forested
 - Jim Corbett N.P.
 - Kaziranga N.P.
- } part of Tarai

Alluvial plains

old Alluvial

Bhangar

New Alluvial

khadar

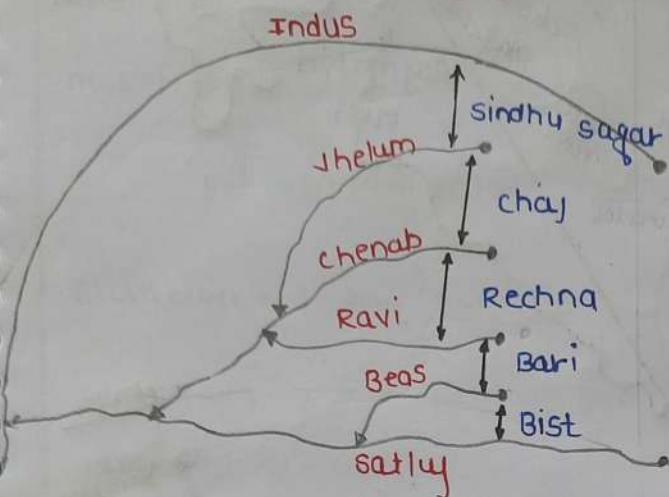
Bhangar

- old Alluvium Flood plains
- Dry & uplifted land
- Kankar found
- Barren
- Not very fertile

khadar

- formed - bank of River
Ganga & Yamuna
- New Alluvium deposits
- most fertile plains
- Local Names
 - Punjab - Dhayyas
 - UP - Khol
 - Bengal - Bhills
 - Bihar - Tidal.

Ooab



- sindhu sagar (N)
- chaj
- Rechna
- Bari
- Bist (S)

davar formations

used for the development
of tea gardens.

*peninsula mostly consists
of relict & residual mount.

- Aravali hills
- Nallamala hills
- Javadi hills
- Veliconda hills
- palconda range
- mahendragiri hills.

Peninsular India

① plateaus

② Hills

plateaus

- mar-war plateau - east of Rajasthan.
↓
West-Aravali

- Bundelkhand plateau - UP + MP

- Baghelkhand plateau - MP
Ken, detwa

- Hazaribagh plateau - Jharkhand
↓
North of Ranchi

- palamu plateau - N-W Jharkhand.

- coimbatore plateau

↓
blw Nilgiris & Dharmapuri
districts.

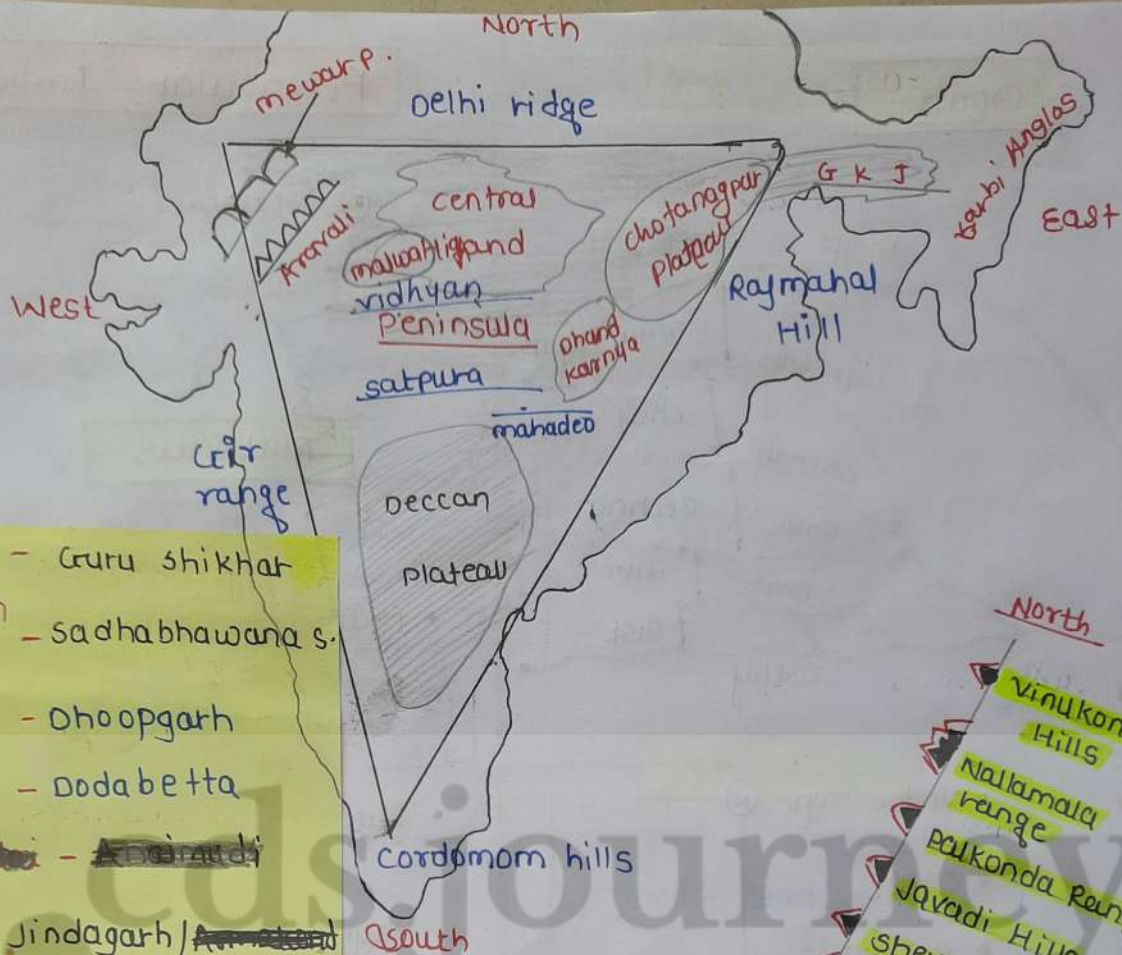
- chotanagpur plateau

↓
famous - coal field

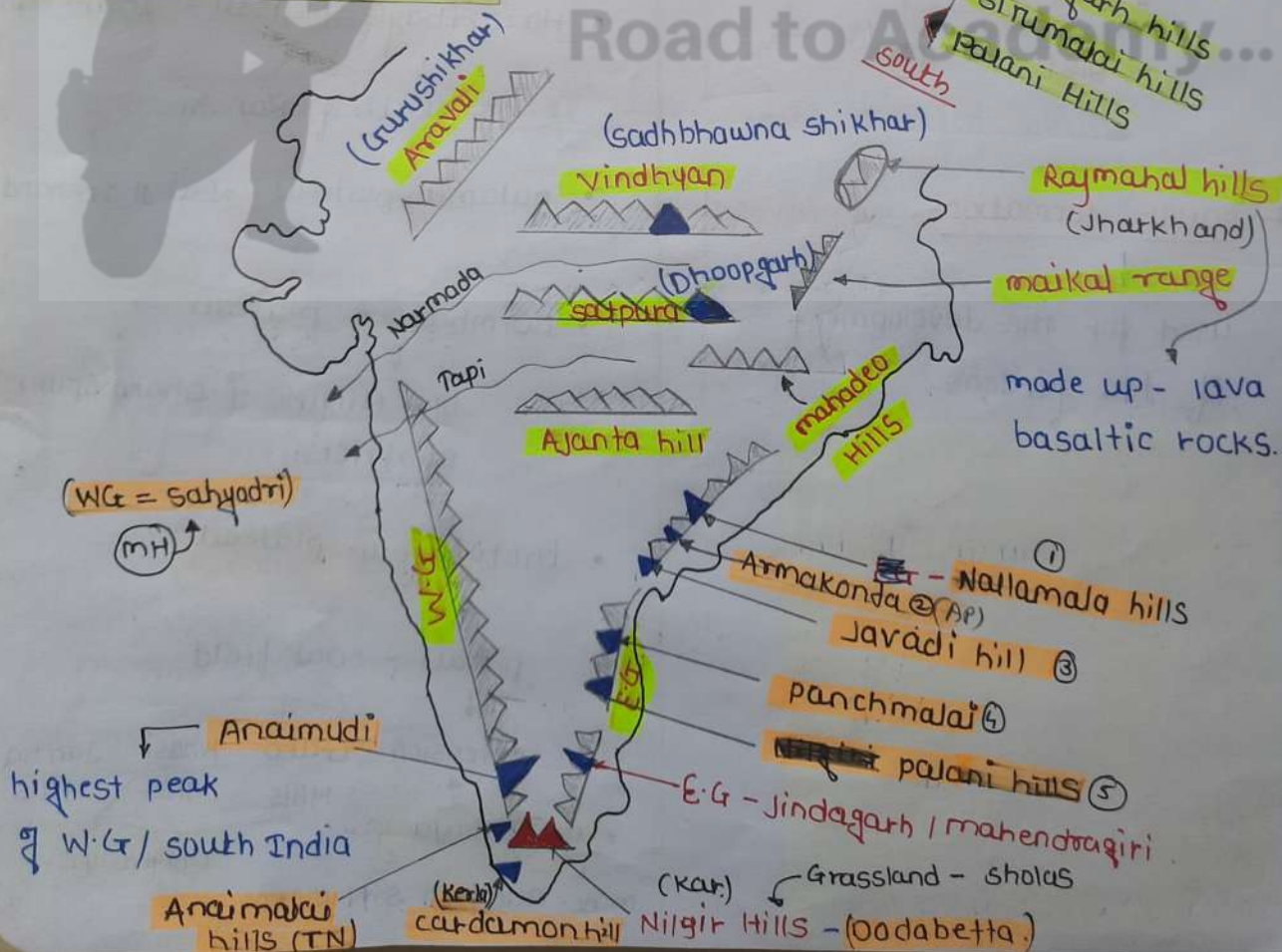
↓
extension - Garo Hills Khasi Hills Jaintia Hills

- meghalaya p. → meghalayas.

max. rainfall S-W monsoon.



- Aravali - Guru shikhar
- Vindhyan Range - Sadhabhawana S.
- Satpura - Dhoopgarh
- Nilgiri - Dodabetta
- ~~Anaimudi~~ - Anaimudi
- E.G. - Jindaghar / Anaimudi
- * Nilgiri hills - E.G. & WG meet



Aravali

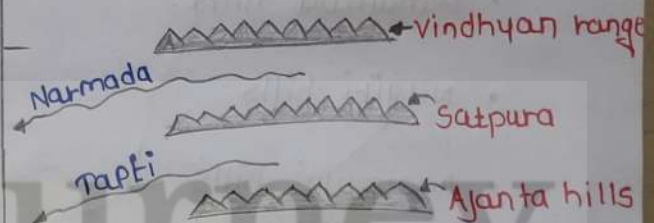
- peak - Guru shikar
- oldest Fold mountain ranges
↓
residual mountain
- Elevation - ↓ S-W ⊕ N-E
- Abu hills
- Jarga hills - udaipur
- marwa - Todgarh
- Harshnath hills - Alwar
- Raisiana hills - Delhi
- Gorang pass - separate
mt. Abu & Guru shik.

Vindhyan range

- peak - shadhabhawana shikar
- Separates - South India & North India
- extension - Gujarat to Bihar
- Sources of -
 - Chambal
 - Betwa
 - Kali sind
 - Ken Rivers
- South - Narmada Flow

Satpura

- peak - Dooptgarh
- Hills
 - Raypipla - GJ.
 - Grawilgarh - MH
 - mahadeo - MP
 - maikal - chh. } extension

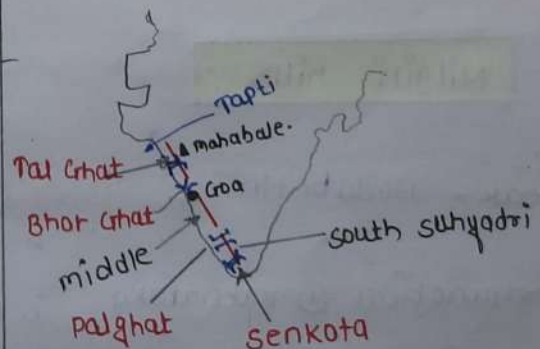


- * Amarkantak plateau

Source of River

- Narmada River
- son River

Western Ghats



- UNESCO World herit. site
- MH - Sahyadri

* Western Ghats - passes

- Thalghat
- Bhorghat
- Palghat
- Senkora



Q. Arrange North (to) South

- Nallamala hills
- Nilgiri hills
- Anaimalai hills
- Cardamom hills

* Western Ghats N to S

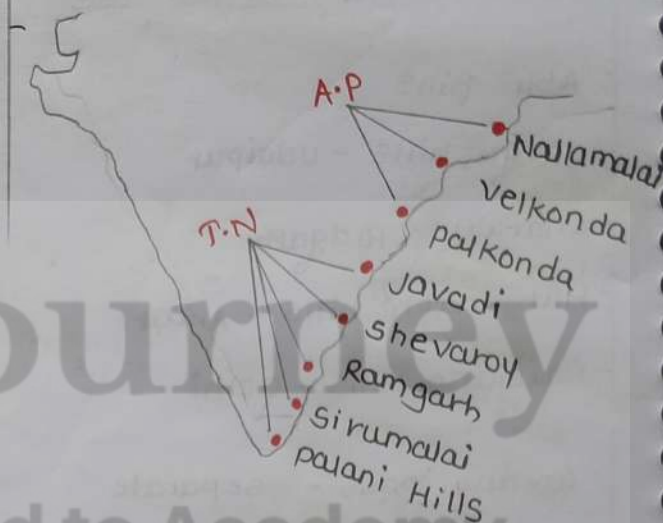
- Kasara Ghats - connect Nasik to Mumbai
- Palakkad Gap
- Malshes Ghats
- Kumbhari Ghats

Nilgiri hills

- peak - Dodabetta
- trijunction ① Karnataka
② Kerala
③ T.N
- separat - W.G + E.G

Eastern Ghats

- discontinuous
- low hills
- peak - Jindagath (AP)



Coastal plains

- coastline - 7500km
- coastal state:
 - ① Guj.
 - ② MH
 - ③ Goa
 - ④ Ktk.
 - ⑤ Ker.
 - ⑥ T.N
 - ⑦ A.P
 - ⑧ Odisha.
 - ⑨ W.B
- ① Df Nagar haveli & Dfd
- ② puducherry.
- coastline:
 - ① Guj.
 - ② T.N
 - ③ A.P
 - ④ MH
 - ⑤ Kerala
 - ⑥ Odisha
 - ⑦ Karnataka
 - ⑧ WB
 - ⑨ Goa.

Rann of Kutch

Kathiawar

Diu

Daman

Konkan Coast

MH

Kar + Ker

Malabar Coast

Subarnrekha River

Chilikalake

odi

Utkal Coast

Northern Circars

Cormandal Coast

famous - Kayals (backwaters)

used

① Fishing

② Inland Navigation

③ attraction for tourists.

* Nehru Trophy Vallamkali (boat race)
held - Punnarnada Kayal (Kerala)

* longest coastlines (India)

① Gujarat

⑧ WB

② T.N

⑨ Goa

③ A.P

④ MH

⑤ Kerala

⑥ Odisha

⑦ Karnataka

Ports

Deendayal
port

(Guj) Porbandar

(MH) Jawahar Lal
Nehru port

Goa
 Panaji
 Mormugao

(Ktk) Mangaluru

Kerala
 Kochi
 Alappuzha

Nagapattinam
 Tuticorin

Haldia (WB)

Paradip (Odisha)

Visakhapatnam

Kakinada

Machilipatnam

Krishnapatnam

Kattupalli

Ennore

Chennai

T.N

Andaman & Nicobar

(highest peak A&N)

Saddle peak

①

14° degree
N



North Andaman

Mt. Diavalo

③

Middle And.



South Andaman

Mt. Koyab

④

Port Blair

Duncan
passage



Little Andaman

10° line



Car Nicobar



Little N.

Mt. Thuller

②



Great N.

6° 45' N

Order high to low

① saddle peak

② mt. Thuller

③ mt. Diavalo

④ mt. Koyab

* Barren Island → Nicobar



Active volcano.

*

blw → 6° N - 14° N

→ 92° E - 94° E

DRAINAGE SYSTEM



* **catchment Area** - River drains the water collected from a specific Area.

River basin

Watershed

large River

small Rivers

Indian River Systems

Himalayan

- origin - Himalaya.

- longer (perennial)

- High Water Volume.

- youth stage

- form - V-shaped valleys

rapids

waterfalls

erosional

ox-bow lakes

flood plains

braided channels

deltas

depositional

Peninsular

- origin - plateau & hills

- shorter (seasonal)

- Low Water Volume.

- old stage

- depends on monsoon

- form • broad

• shallow valleys

* East Flowing

77% of Indian drainage
(BOB)

- Ganga
- Brahmaputra
- Mahanadi
- Godavari
- Krishna

West Flowing

23% of Indian Drainage
(Arabian sea)

- Narmada
- Tapi
- Mahi
- Sabarmati
- Periyar

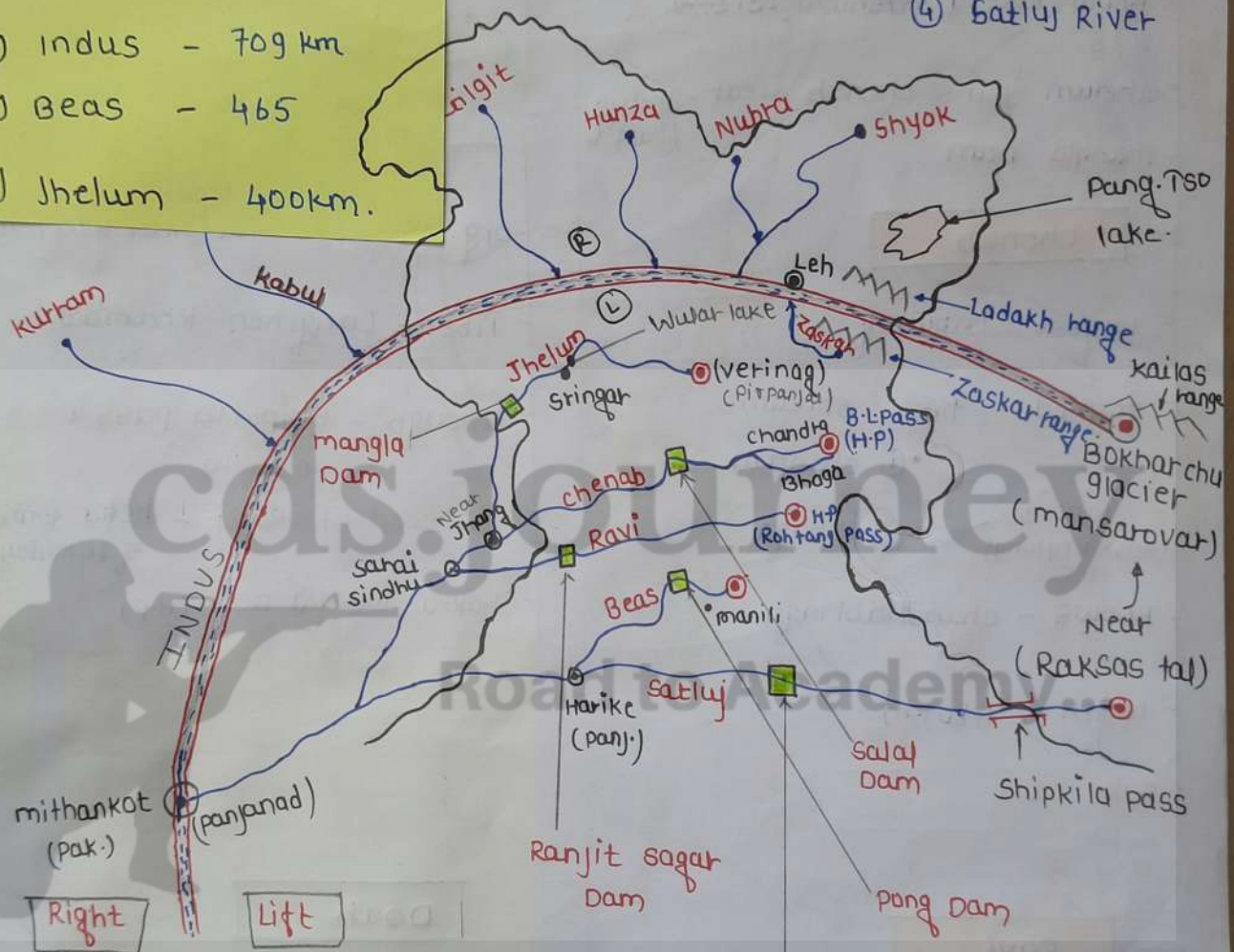
The Indus System

↓ Descending order.

- 1) Chenab - 1180 km
- 2) Satluj - 1080 km
- 3) Ravi - 725 km
- 4) Indus - 709 km
- 5) Beas - 465
- 6) Jhelum - 400 km.

④ North to south

- ① Shyok River
- ② Zaskar River
- ③ Spiti River
- ④ Batluj River



Indus

- Originate - Bokhar Chu glacier, at Kailas range (Tibet)
- length - 2880 km
- Tibet - Singi Khamban
- Ind + Pak + Afg + Tibet

Basin

- J+K
- Ladakh
- Punjab
- Rajasthan
- Haryana
- Chandigarh
- H.P

- * Satluj
↓
longest tribu.
- * Chenab
↓
largest trib.

Jhelum

- ori. - Verinag (J+K, Pir Panjal)
- city - Srinagar • Uri • Baramulla
- Wular lake (freshwater lake)
- Jhelum joins Chenab Near - Jhang (Pak)
- mangla dam

Chenab

- largest tributary
- formed - two streams
 - ① chandra
 - ② bhaga
- Tandī (near Keylong) H.P.
- knows - Chandrabhaga
- length - 1180 km
- Sakai dam

Ravi

- ori. - Rohatung Pass (H.P.)
- join Chenab in (Pak) - sarai Sindh
- Ranjit Sagar dam
- city - Amritsar
- Pathankot

Beas

- ori. - Rohtang pass - Beas Kund
- joins - Satluj at Harike
- pong dam

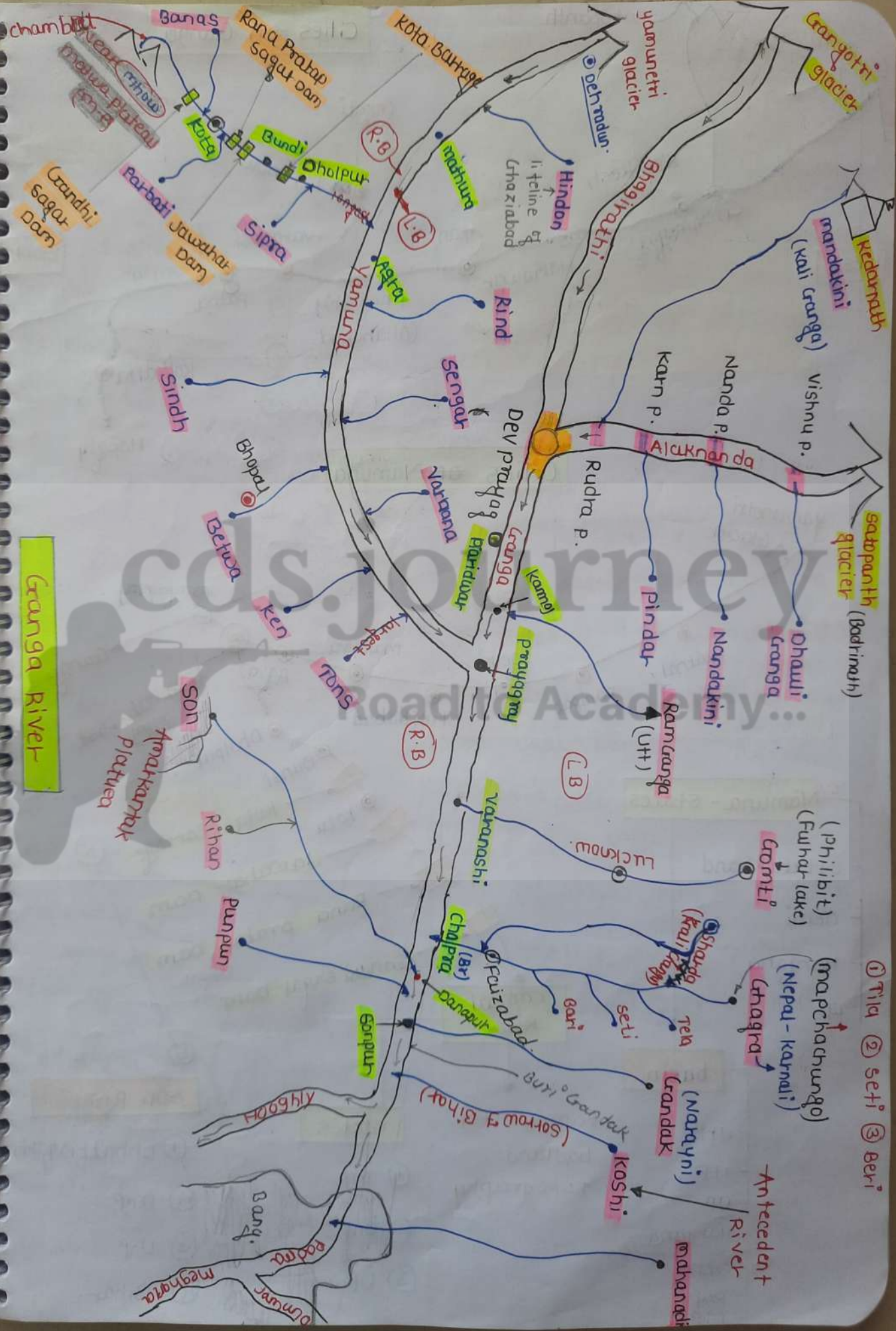
Satluj

- ori. - Rakas tal, Near Mansarovar
- Tibet - Langchen Khambab
- through - Shipkila Pass inter in India.
- Antecedent River - before formation of Himalaya.
- Bhakra Nangal Dam (H.P.)

old Name

- ① Indus - Sindhu
- ② Jhelum - Vitastata
- ③ Chenab - Askini
- ④ Ravi - Purushni
- ⑤ Beas - Vipasha
- ⑥ Satluj - Shatadri

Chenab	}	Chay
		Rechna
Ravi	}	Bah
Beas		
Satluj	}	Bist



Ganga River

- ① Tila ② seti ③ Beri

(mapachungo)
(Nepal - Karnali)
Ghaghara

(Philibit)
(Fulhar lake)
Gomti

(Sharda)
(Kailash)
Teta

(Naryani)
Gandak
Koshi

Antecedent River

mahanadi

Lucknow

(L.B)

Rudra P.

Hindon

Li teline of
Gha zabad

Rind

Seagor

Varana

Prayagraj

Varanashi

Chhapra

Patna

Gorakhpur

Baran

Jaunpur

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

(L.B)

(R.B)

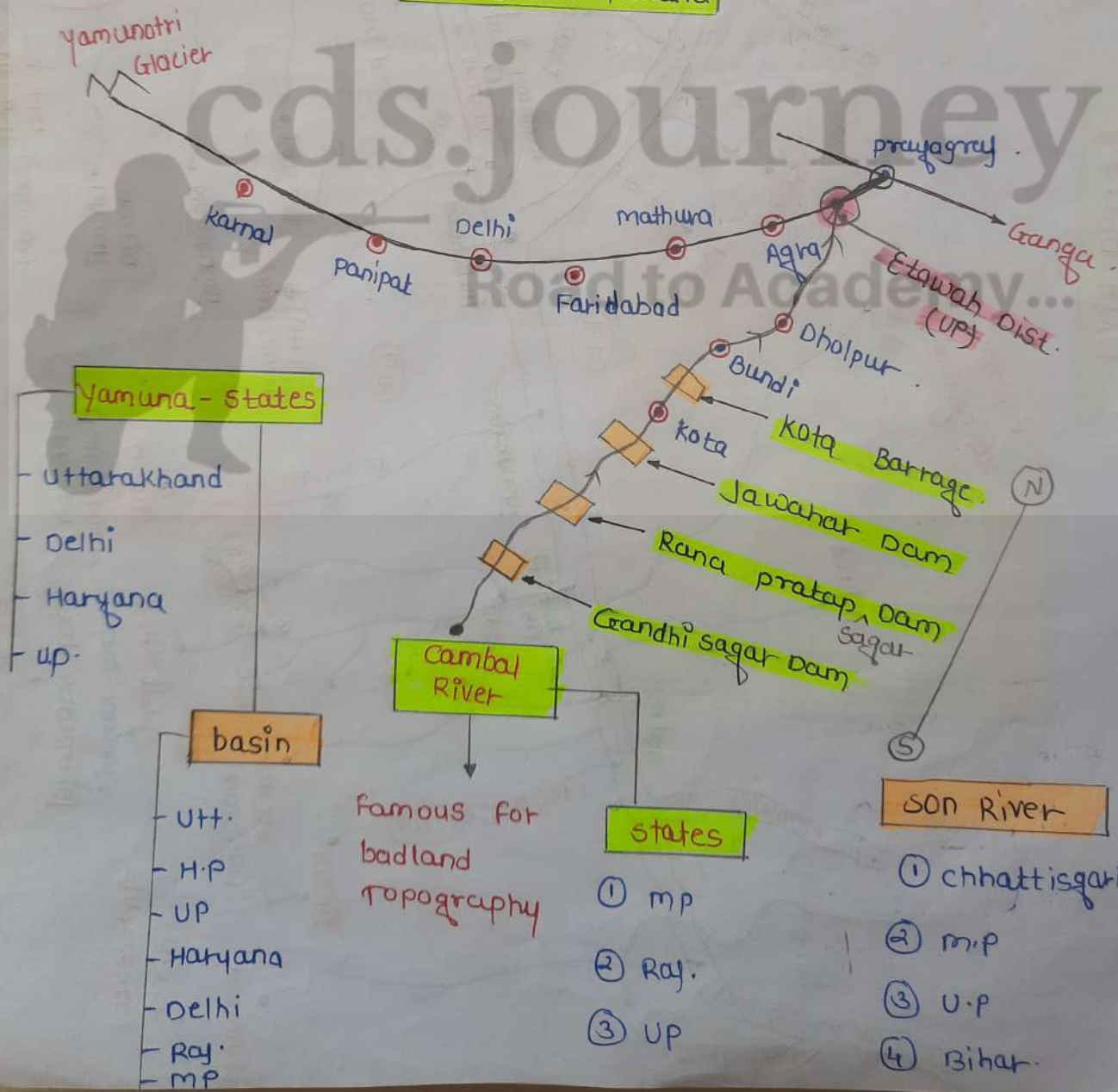
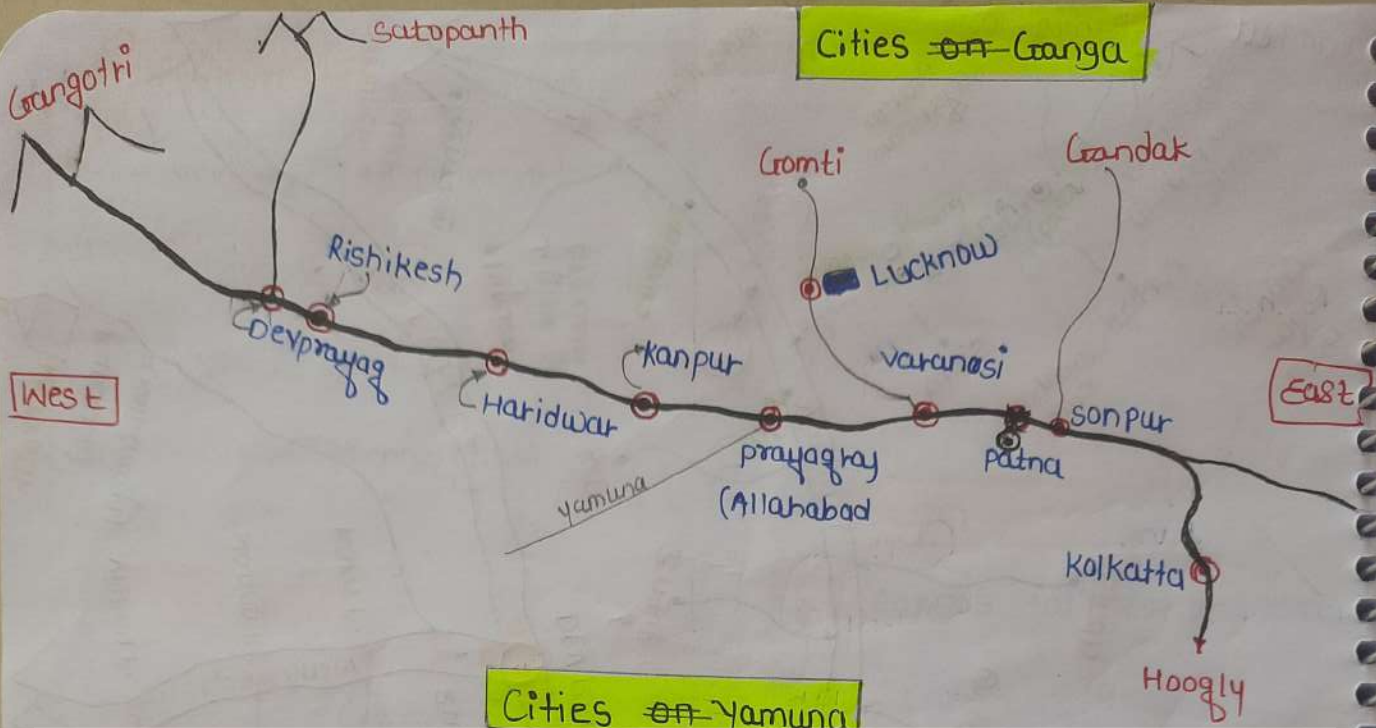
(L.B)

(R.B)

(L.B)

(R.B)

(L.B)



Ganga

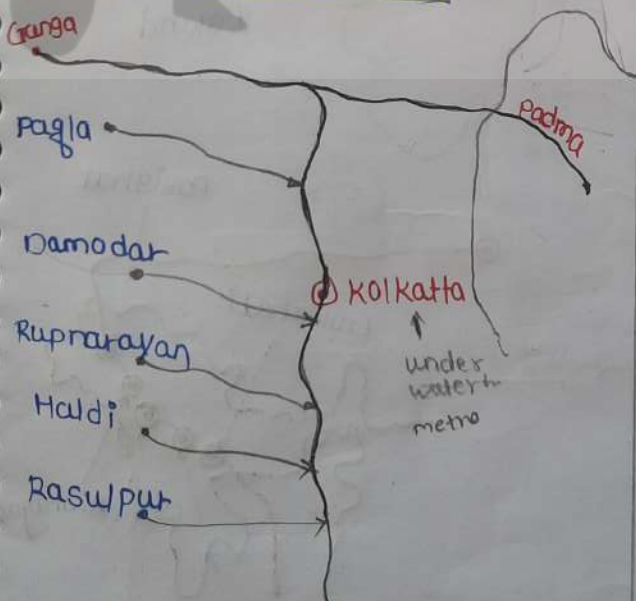
State

- ① Uta.
- ② UP - ↑ Bijnor Dist.
- ③ Bihar
- ④ Jharkhand
- ⑤ W.B

Basin

- ① Uta.
- ② UP
- ③ Haryana
- ④ Himachal P.
- ⑤ Delhi
- ⑥ Bihar
- ⑦ Jharkhand
- ⑧ Rajasthan
- ⑨ mp
- ⑩ chhatti.
- ⑪ W.B.

Hooghly River



- 1) Dudhsagar Falls - Goa
- 2) Thalaiyar Falls - T.N
- 3) Barkana Falls - Ktk
- 4) Jog Falls - Ktk
- 5) Kunchikal Falls - Ktk
- 6) Barehipani Falls - Odisha
- 7) Mawsmai Falls - Meghalaya
- 8) Thalaiyar Falls - T.N.

Dam

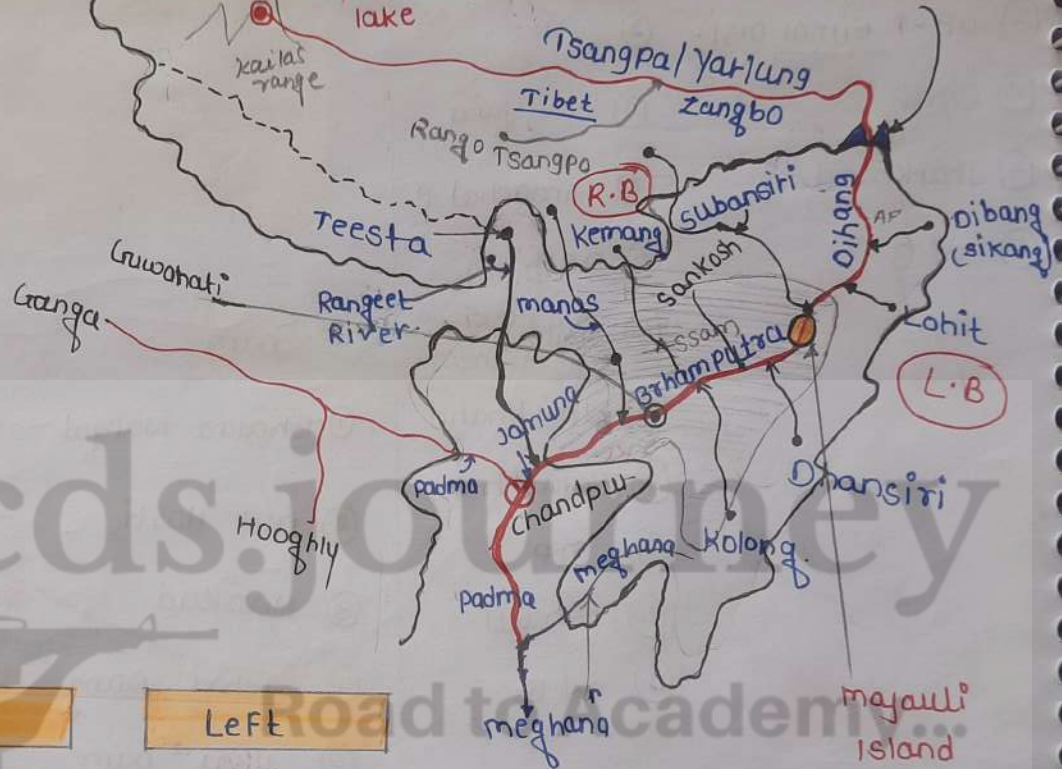
River

- ① Bhakra Nangal — Satlej
- ② Dw Hasti — chandra
- ③ Hirakud — Mahanadi
- ④ Tehri Dam — Bhagirathi
- ⑤ Ukai Dam — Tapi
- ⑥ Krishnaraja Sagar Dam — Kaveri (Ktk)
- ⑦ Srisailem Dam — Krishna
- ⑧ Mettur Dam — Kaveri (TN)
- ⑨ Cheruthoni Dam — Kerala (same)
- ⑩ Indira Sagar Dam — Narmada (mp)
- ⑪ Sardar Sarovar Dam — Narmada

india - 915km

chemayungdung Glacier
(Near Mansarovar)
lake

enter- Namcha
Barva.



Left

- ① Dibang / sikang
- ② Lohit
- ③ Phansiri
- ④ Kolong

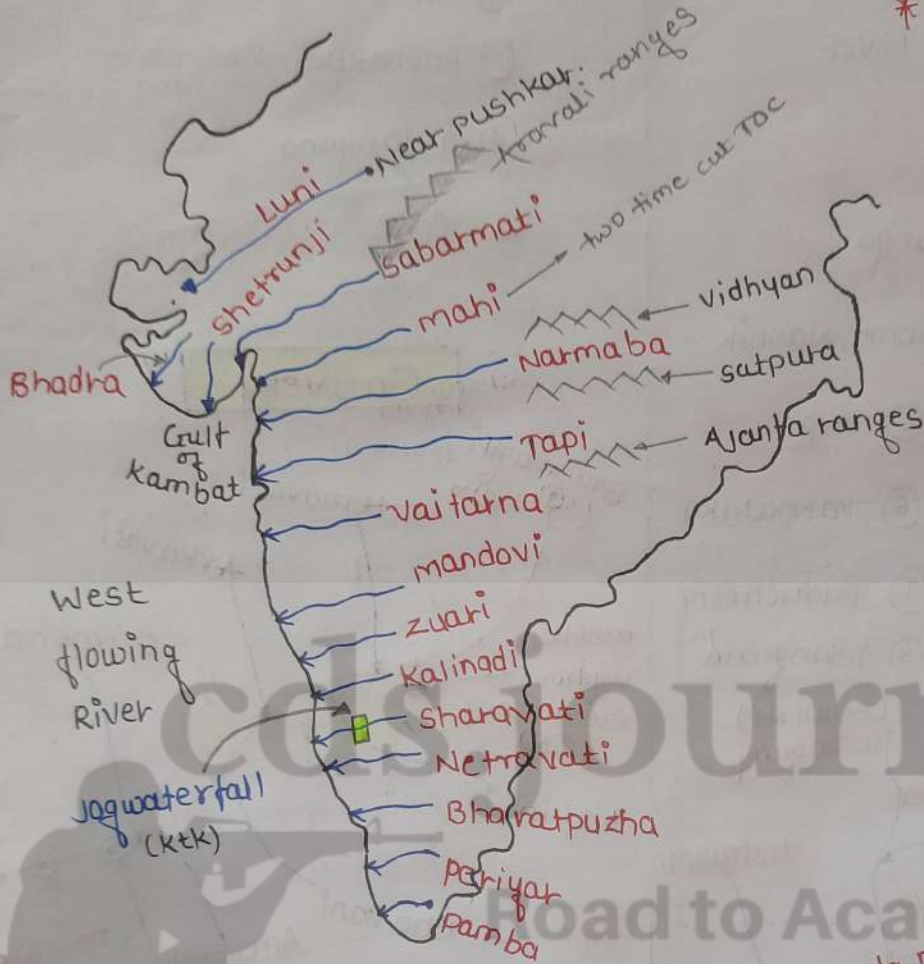
Mayaumi Island
↓
World largest River Island



Peninsular River

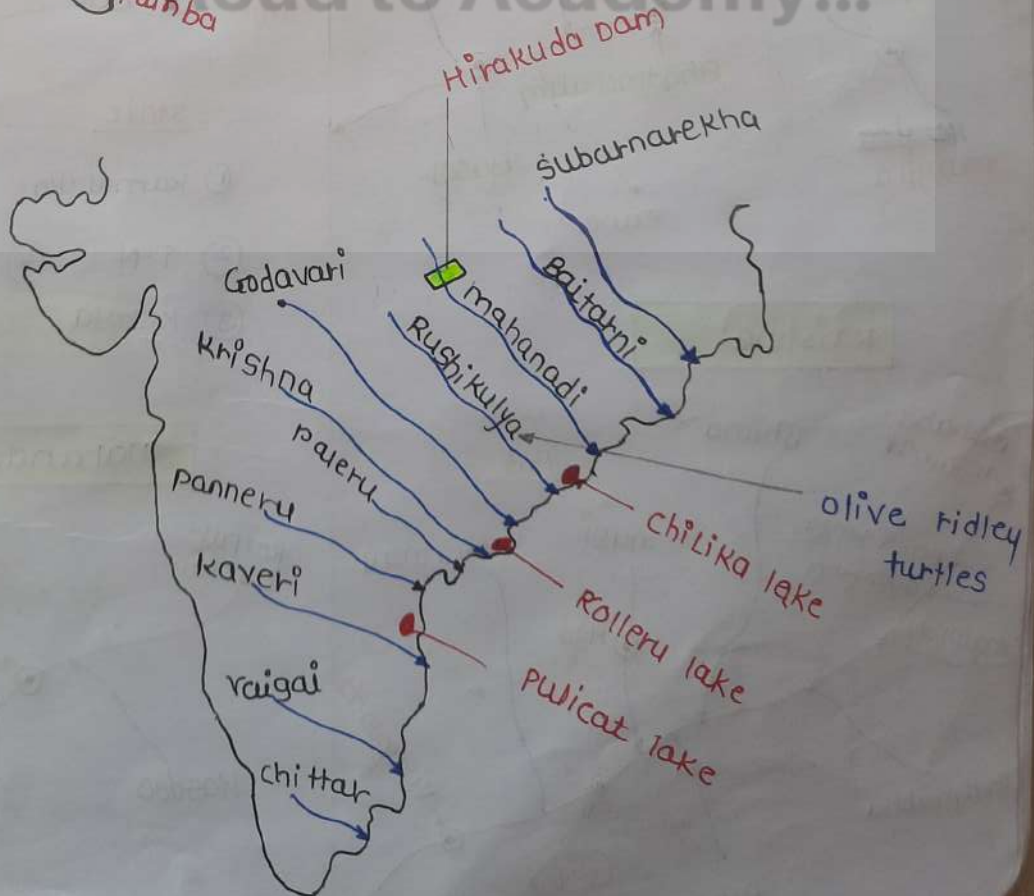
* tributary of River Luni
[cos 2018]

- ① Khari
- ② Sukri
- ③ Jawai



West flowing River

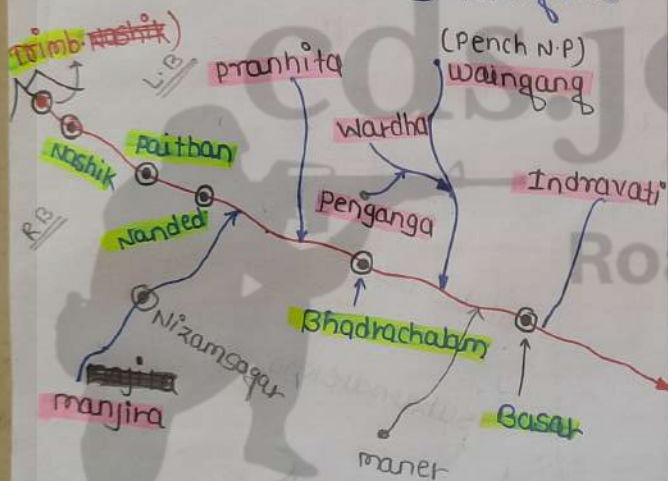
Jog water fall (Krk)



Cradavari

- largest peninsular river
- 2nd largest basin
- called - Dakshin Ganga
- rise - Trimbakeshwar, Nashik

- basin -
- | | |
|----------|--------------|
| ① MH | ⑤ MP |
| ② A.P | ⑥ Karnataka |
| ③ Chh. | ⑦ Puducherry |
| ④ Odisha | ⑧ Telangana |



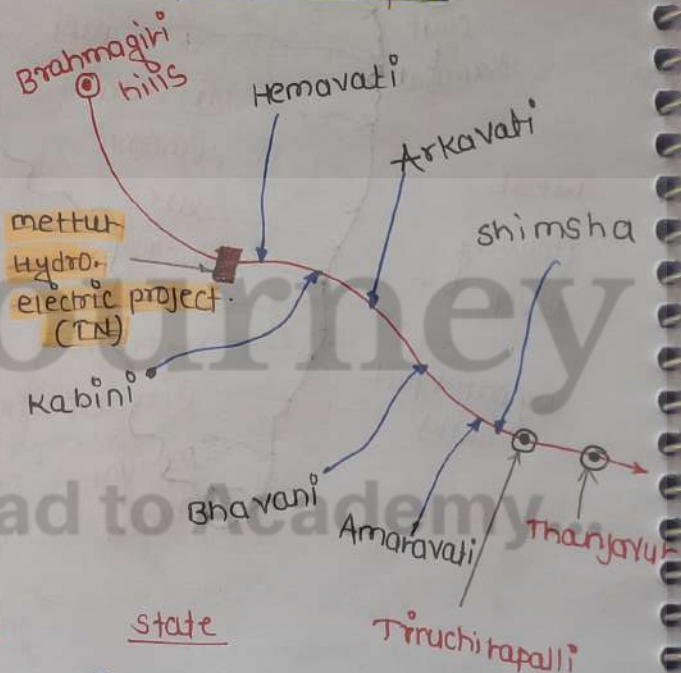
Krishna



states

- ① MH
- ② Karnataka
- ③ Telangana
- ④ A.P

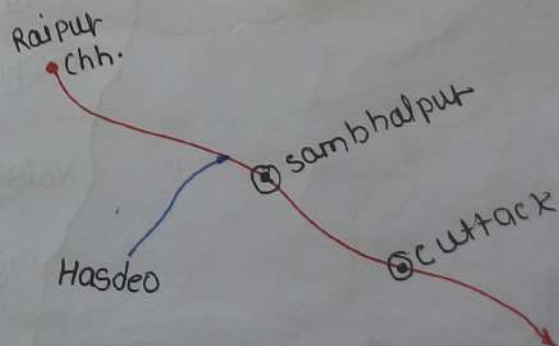
Cauvery



state

- ① Karnataka
- ② T.N
- ③ Kerala
- ④ Puducherry

Mahanadi

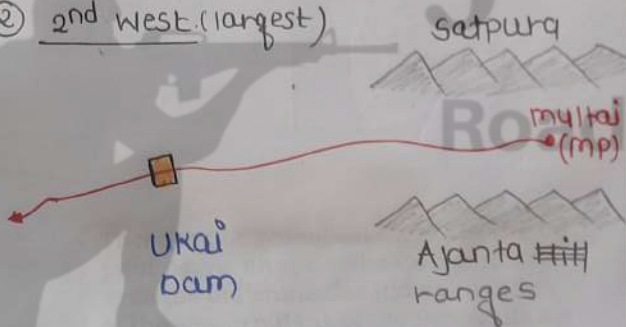


Narmada



Tapi

② 2nd West (largest)



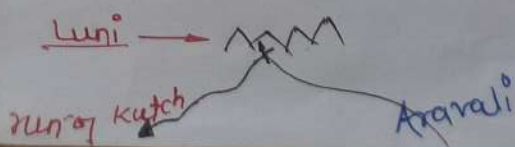
Sabarmati

ori - Aravali range. Raj

① Raj ② Gwy.

water gives - Ukai Dam

Orane - Gulf of Khambhat.



River Originate - Amarkantak plateau

- ① Narmada
 - ② Son
 - ③ Johila
- Radial pattern

length wise Top 10 Rivers

- ① Ganga - 2525 km
- ② Godavari - 1465 km
- ③ Krishna - 1400 km
- ④ Yamuna - 1376 km
- ⑤ Narmada - 1312 km
- ⑥ Indus - 1111 km
- ⑦ Brahmapu. - 916 km
- ⑧ Mahandi - 850 km
- ⑨ Cauvery - 800 km
- ⑩ Tapi - 720 km.

* Total length

Indus } Brahmapu. } Ganga.

- Ayodhya - Sarayu
- Ferozpur - Satlej
- Hyderabad - Musi
- Jabalpur - Narmada
- Jammu - Tawi
- Surat - Tapi
- Mirzapur - Ganga
- Bangalore - Vrishabhavati

Concordant patterns

path of river dependent on
slope & topography

① consequent patterns

Follow - direction of slope
most - peninsular.

• Godavari ...

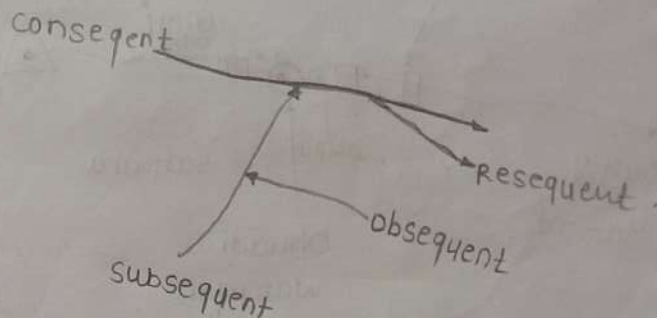
② Subsequent patterns

formed by headward
erosion

- Ex:
- Ken
 - Chambal
 - Sind
 - Betwa
 - Tong

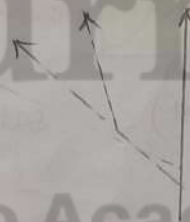
③ obsequent Rivers

form - right angles



④ Resequent Rivers

flows - same direction

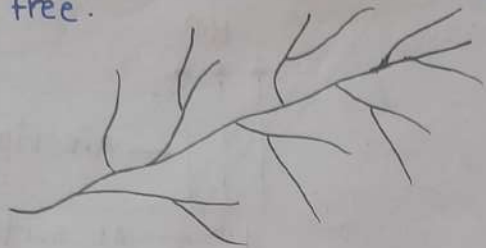


River	Catchment area sq. km
Sabarmati	21,674
Mahli	34,842
Dhandhar	2,770
Kalinadi	5,179
Sharavati	2,029
Bharathapuzha	5,397
Periyar	5,243

River	Catchment area sq. km
Subamarekha	19,296
Battani	12,789
Brahmani	39,033
Penner	55,213
Palar	17,870

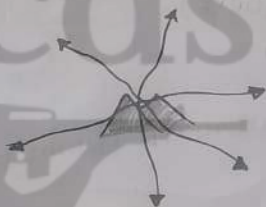
Dendritic

resembling the branches of a tree.



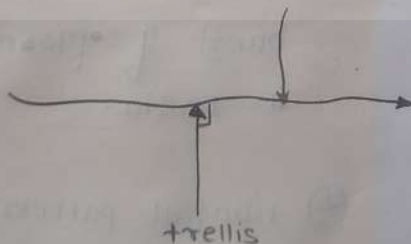
Radial

originate from hill & flow in all direction



trellis

^ join right angle tributaries



Centripetal

River discharge their waters from all directions in a lake or depression



Lake

Type

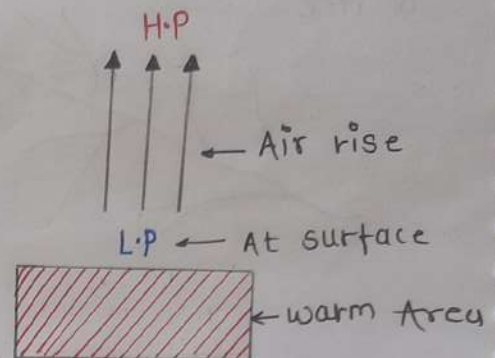
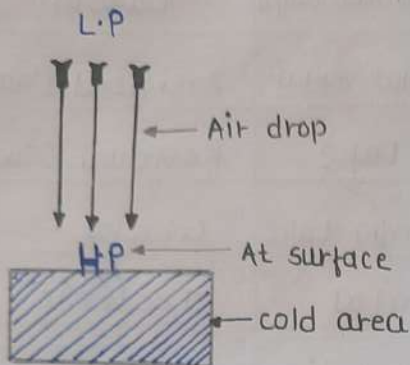
Lake	Type	State
Ashtamudi Kayal	Lagoon	
Himayat sugar	Artificial (sweet)	
Dal Lake	Residual (sweet)	
Tsomo Lake	Glacial	Sikkim
Bhimtal	Tectonic	
Ashamudi	Lagoon	
Cochina	Landslide	
Lonar	crater	
purbasthali	Fluvial	
Trangabai	Glacial	
Kaliveli	Lagoon	T.N
Khajjiar	mid altitude	H.P
Khecheopalri	Sacred	Sikkim
Vembanad		Kerala
Barapani		Megh.
Kolleru		Andhra A.P

① Syama Prasad Mookerjee Tunnel - Chenani - Nashri (JK)
longest tunnel (NH44)

② Jawahar Tunnel - blw ^{sringar} Jammu

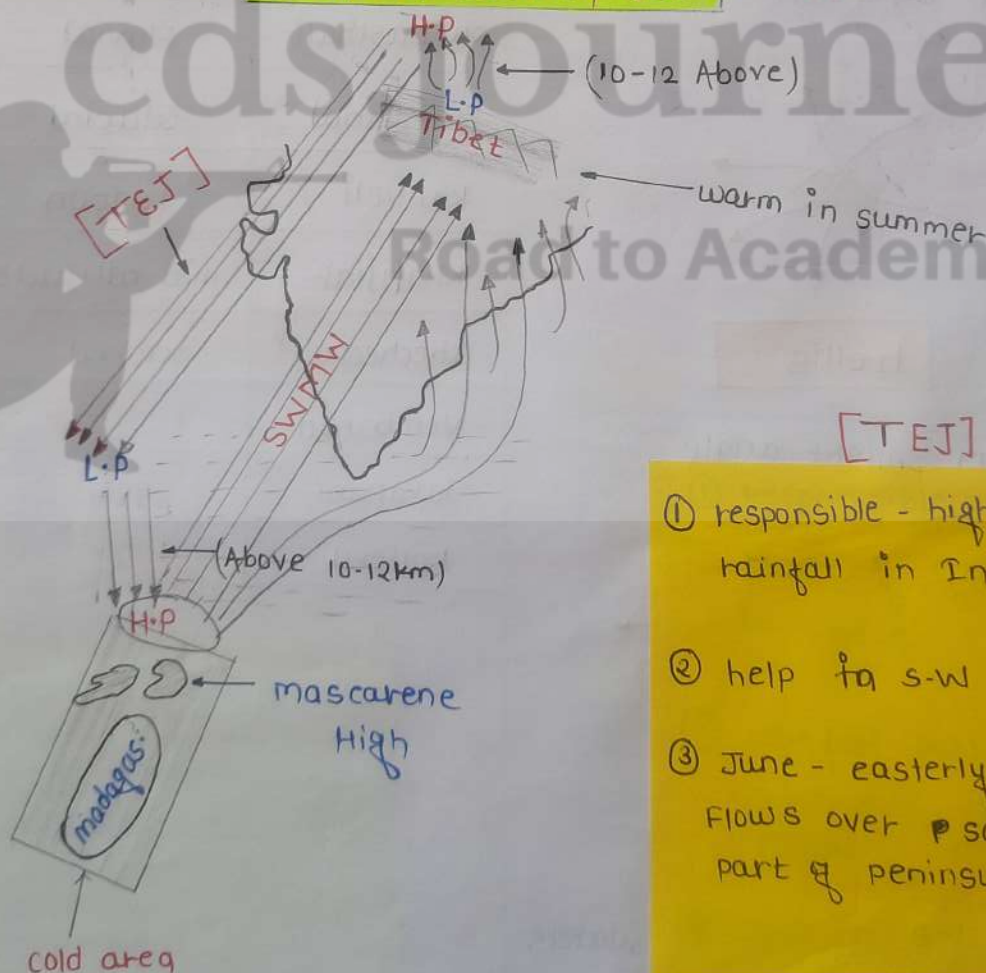
③ Atal tunnel - Manali - Leh

Indian Climatology



Tropical Easterly Jet

Jet - air above 10-12 km



[TEJ]

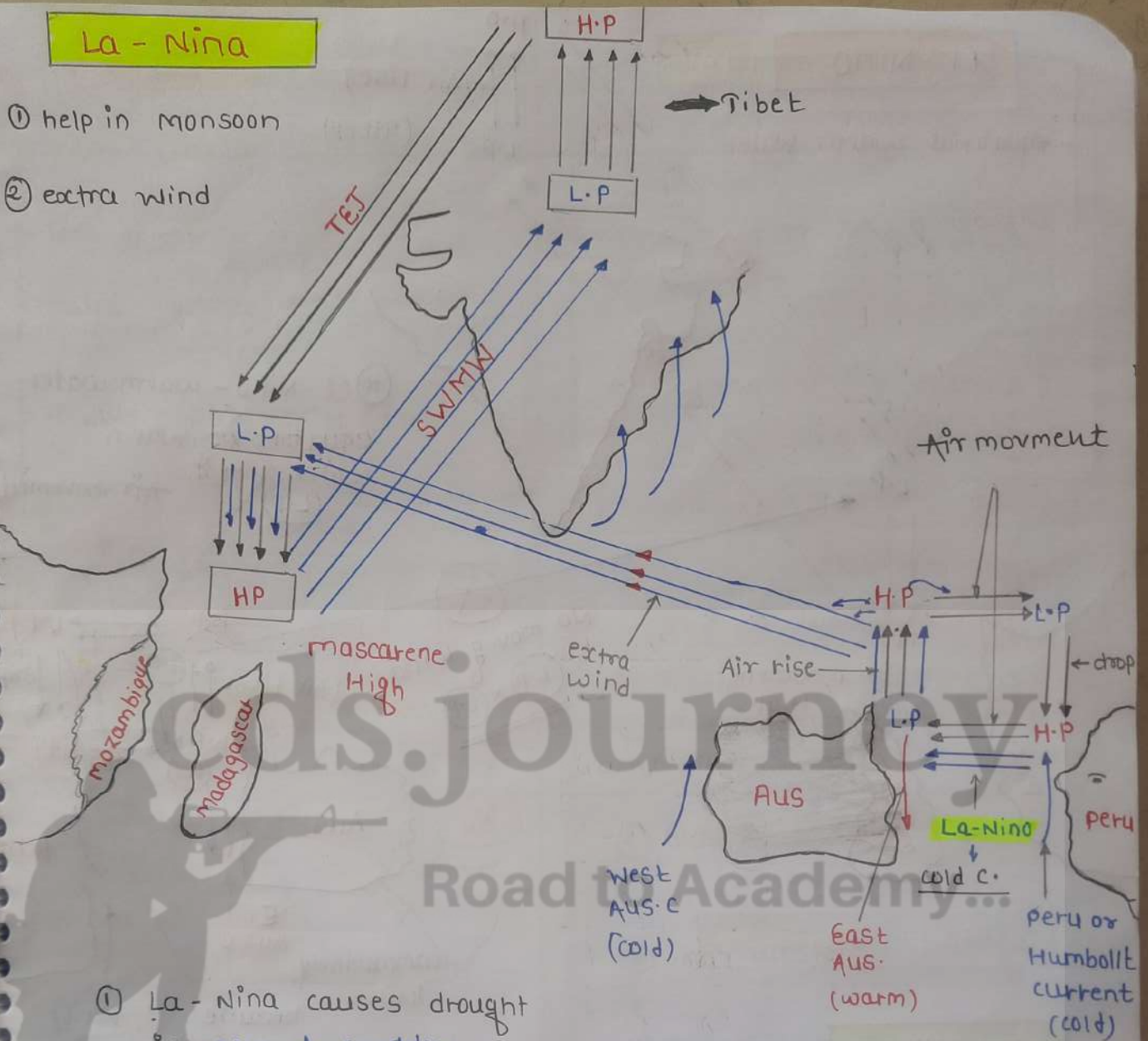
- ① responsible - highest rainfall in India.
- ② help for s-w monsoon
- ③ June - easterly jet stream flows over southern part of peninsula.

cold area
compared to
Tibet

⊕ Somali current - also help in
became monsoon strong
Not every time

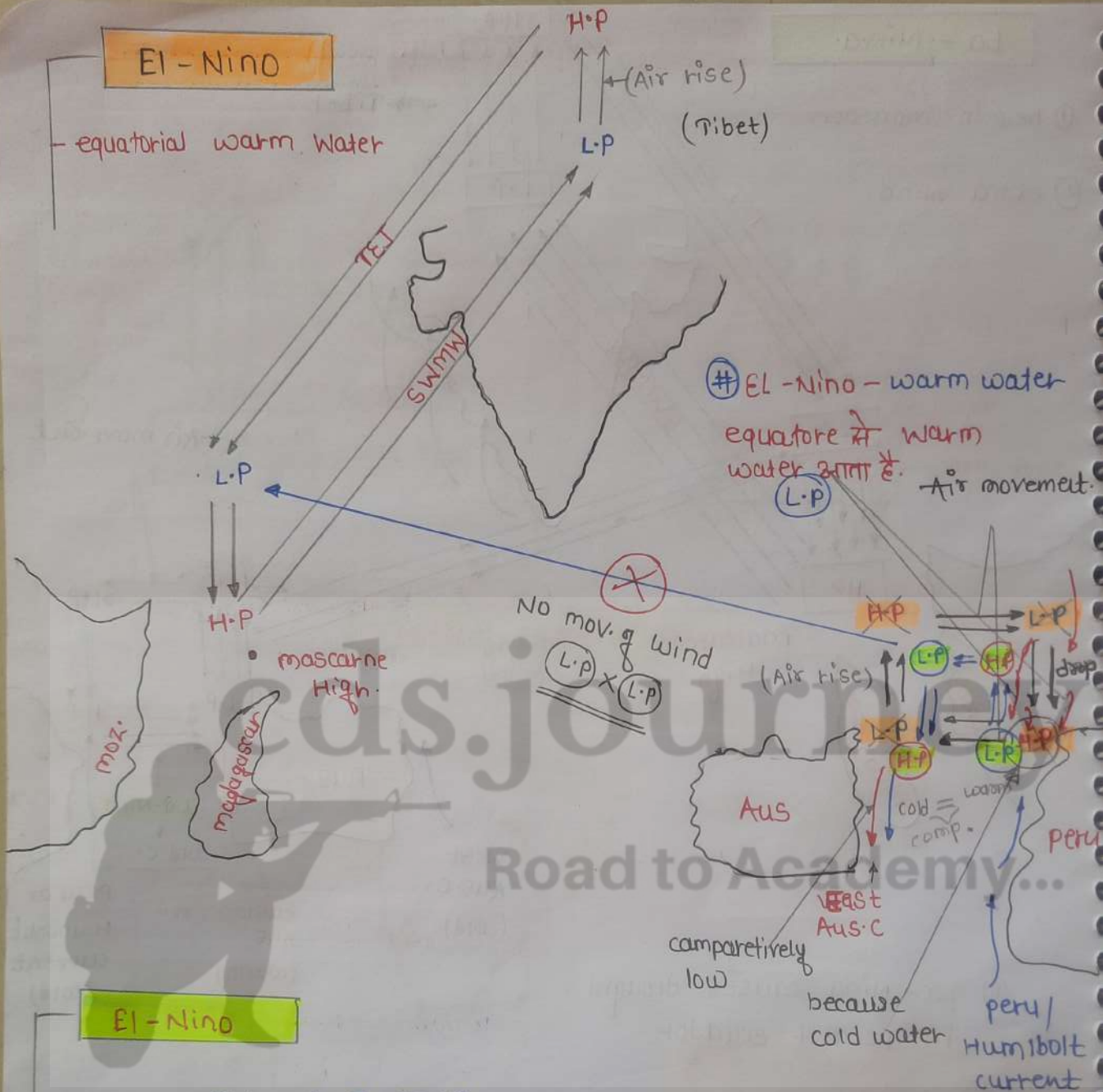
La - Nina

- ① help in monsoon
- ② extra wind



- ① La - Nina causes drought in peru + ecudor
- ② heavy Floods - Aus.
- ③ cold current.

El-Nino



El-Nino

- appears every 3 to 7 years
- Warm equatorial current.
- get replaced temporarily peru (or) Humbolt current.
- ↑ tem. 10°C in peru cost.
- ① distortion of equatorial atmospheric circulation
- ② irregularities in the evaporation of sea water.
- ③ reduction → planktons → reduced fish in the sea.
- used in India for forecasting long range monsoon rainfall.

ITCZ

- Inter Tropical convergence zone.
- low pressure zone
- located - where trade wind converge.
- In July - $20^{\circ}\text{N} - 25^{\circ}\text{N}$ [ITCZ]
- Winter - ITCZ moves southward
 $\downarrow$
 N-SW wind $\rightarrow$ Northeast monsoon

Western disturbance (winter)



* causes

① rainfall in

- Punjab
- Haryana
- Delhi
- Western up.

Mediterranean climate

- Temp. range is moderate with warm to hot summers and mild winters.
- dry summers

Northeast monsoon

- during - Oct. & Nov.
- crossing over bay of Bengal picks up moisture
 $\downarrow$
 causes.
- * rainfall - TN, southern A.P
 south Ktk. & Kerala (SE)

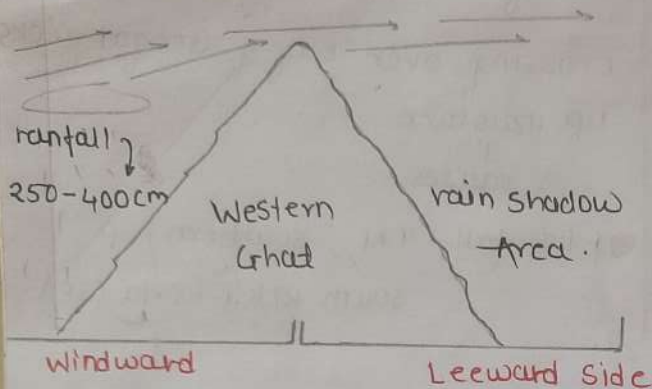
thunderstorms

#

famous local ^ storms

- ① mango shower - Kerala & coast Ktk.
(mangoes)
- ② Blossom shower - Kerala
(coffee)
- ③ Nor Westers - Bengal & Assam
 $\downarrow$
 useful i) tea
 ii) jute
 iii) rice cultivation.
- ④ Assam $\rightarrow$ Bardoli chheerha
- ④ Loo - Panj. to Bihar
- ⑤ cherry Blossom - TN / Karmat.

Arabian branch

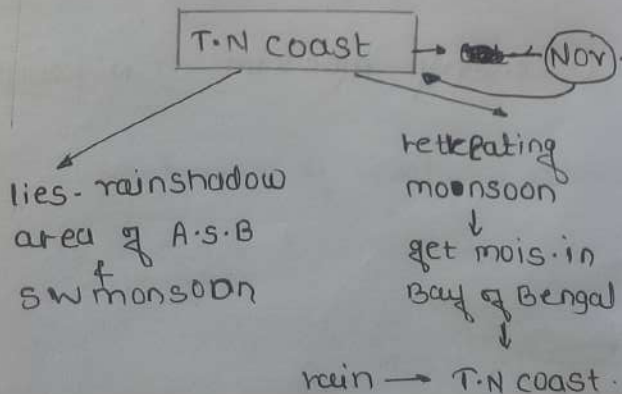


* Arabian sea branch + Bay of Bengal branch meet
↓
Amritsar, Punjab.

Bay of Bengal

Arakan hills along coast of Myanmar deflect toward Indian sub.

high rainfall.



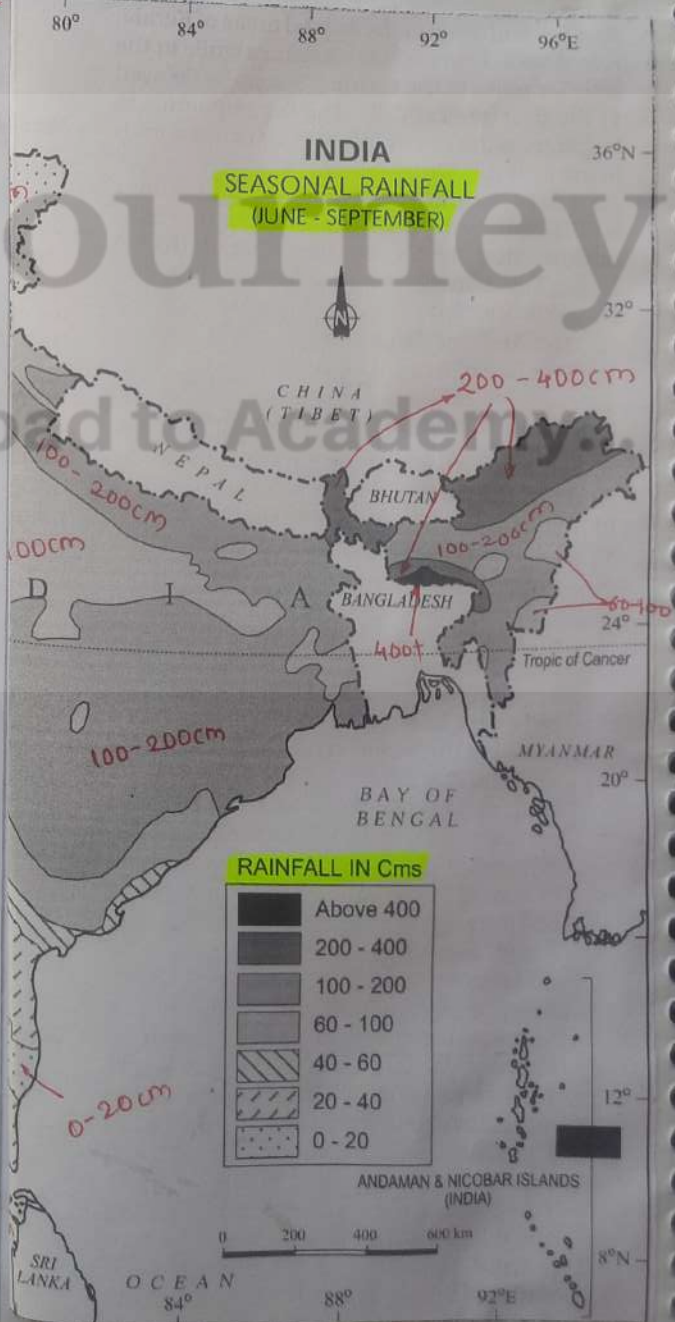
High Rainfall

200cm+ rainfall.

• West coast • WG

• hill of meghalaya

1000cm - Khasi + Jaintia hills





Koeppen's Scheme

5 major climate Group

4- on the basis of temp.

- ① A
- ② B C
- ③ D
- ④ E

1- on the basis of precipitation

- ① B

a b c d → degree of severity

A - Tropic	Avg. temp → coldest month → 18°C or higher
B - Dry climates	evap. > precipitation.
C - Warm temperate	Avg. temp → coldest month -3°C to 18°C
Taiga D - cold snow Forest climates	Avg. temp → " " → -3°C or less
Tundra E - High Land	cold due to elevation. (all month) → below 10°C

(dry & hot)

h - Tropical

K - Temperate

f	no dry season
m	monsoon climate
w	winter dry season
s	summer dry

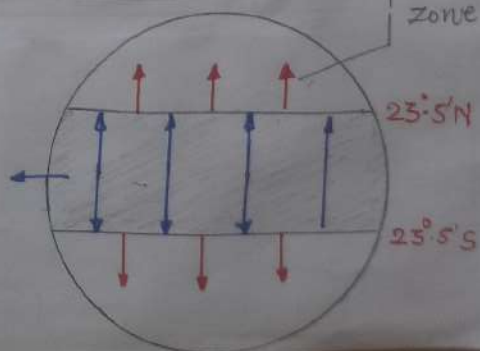
⊕ ET - No true summer

⊕ EF - perennial ice.

B-climate sys.

S (semi Arid) W (deserts) (Arid)

Tropical Zone



Group	Type	Letter Code	Characteristics
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSH	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Tundra climate (ET)

- region of permafrost - sub soil permanently frozen.
- polar region
- summer - very long duration day light.
- No tree in Tundra climate
- North of Arctic circle
- south of Antar. circle
- low growing found
 - ① mosses
 - ② lichens
 - ③ flowering plants.
 - ④ cotton grass
 - ⑤ sedges
 - ⑥ willows

Ice cap climate (EF)

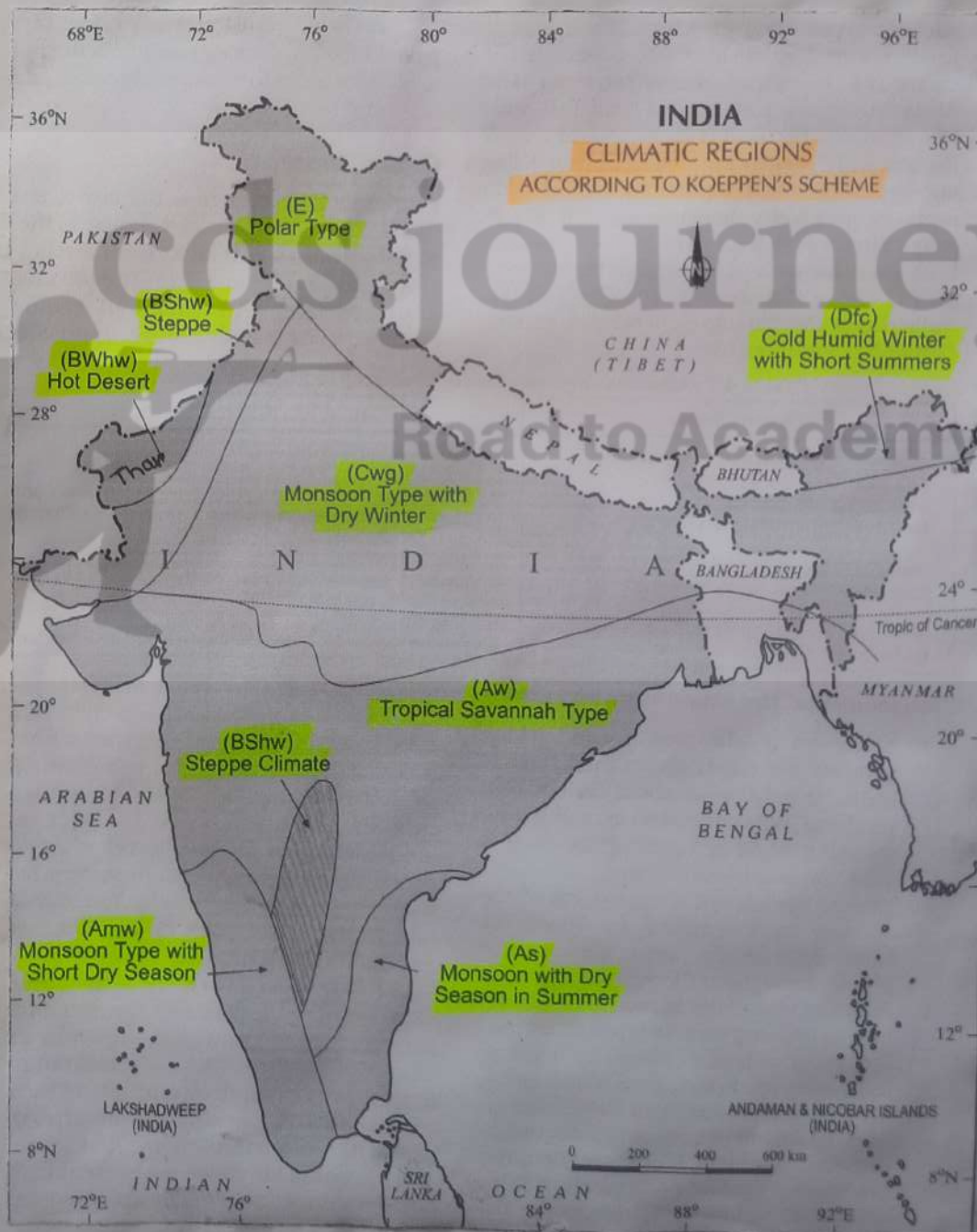
- Greenland & Antarctica
- Temp. always - below freezing point (0°C).

• countries with tundra climate

- ① Alaska
- ② Northern Canada
- ③ Greenland
- ④ Iceland
- ⑤ Northern Scandinavia
- ⑥ Northern Russia
- ⑦ Antarctica.

* Cwg climate → Create Northern plains of India.

Type of Climate	Areas
Amw Monsoon with short dry season	West coast of India south of Goa
As - Monsoon with dry summer	Coromandel coast of Tamil Nadu
Aw - Tropical savannah	Most of the Peninsular plateaus, south of the Tropic of Cancer
Bshw - Semi-arid steppe climate	North-western Gujarat, some parts of western Rajasthan and Punjab
Bwhw - Hot desert	Extreme western Rajasthan
Cwg - Monsoon with dry winter	Ganga plain, eastern Rajasthan, northern Madhya Pradesh, most of North-east India
Dfc - Cold humid winter with short summer	Arunachal Pradesh
E - Polar type	Jammu and Kashmir, Himachal Pradesh and Uttarakhand



Taiga

- ① denote - D
- ② Temp $\rightarrow -10^{\circ}\text{C}$
- ③ snow
- ④ plants $\checkmark$
- ⑤ away from pole
- ⑥ Animals $\checkmark$

Tundra

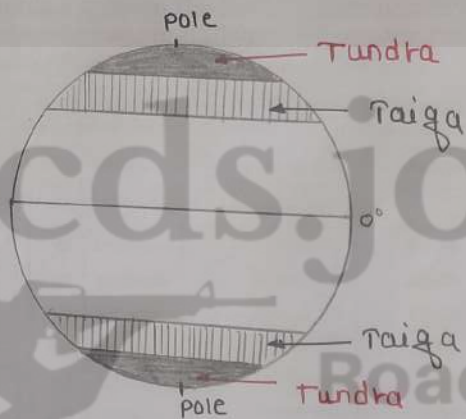
- ① denote - E
- ② Temp $\rightarrow -50^{\circ}\text{C}$
- ③ frost
- ④ plants (X)
- ⑤ at pole
- ⑥ Animals $\checkmark$
- ⑦ rainfall - $\downarrow 30\text{cm}$

- rainfall - 80 - 160 cm

- rainfall $\downarrow$ With distance from eq.

- Temp. - $18^{\circ}\text{C} - 30^{\circ}\text{C}$

→ mosses & lichens from part of tundra vegetation



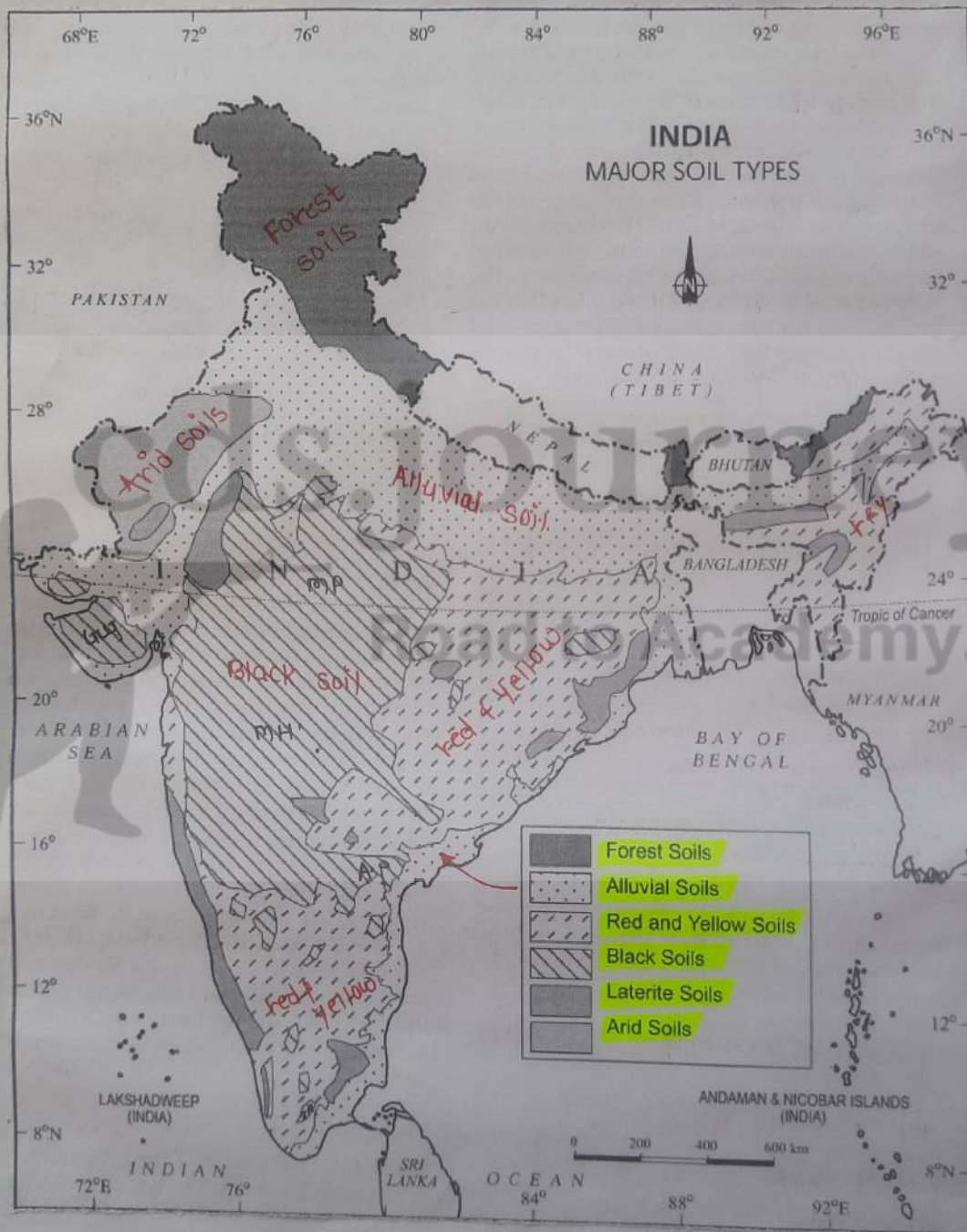
Savanna climate

- also known - sudan climate
- tropical wet & dry climate
- btw - eq. forests & trade wind deserts.
- two seasons
 - ① winter
 - ② summer

* monsoon *
- largest savanna - located in Africa. near equator.

SOIL

5. Dec. World Soil day



* Shankaracharya hills - Srinagar

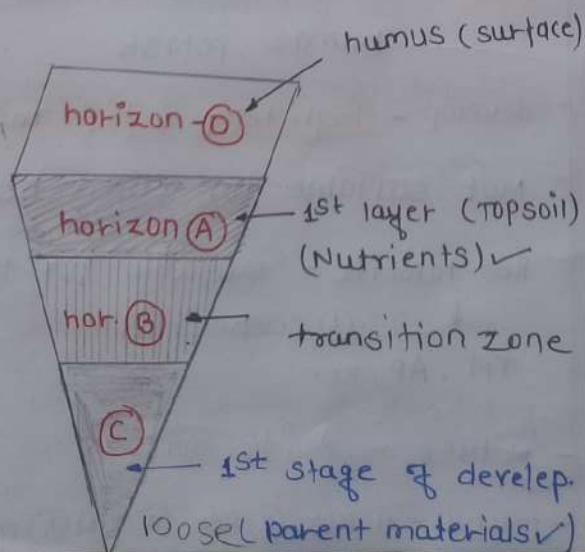
* **sandy** - bigger particles of soil

* **clayey** - fine particles of soil
(water holding capacity very good)

* **Loamy** - Amount of large & small particles are same.

Indian Council of Agricultural Research [ICAR] has classified Indian soil on the basis of their Nature & character as per [USDA] soil taxonomy.

Sl NO	order	%
1	Inceptisols	39.74
2	Entisols	28.08
3	Alfisols	13.55
4	Vertisols	8.52
5	Aridisols	4.28
6	Ultisols	2.51
7	Mollisols	0.40
	others	2.92



Soil depends on factors

* parent material

* climate

* topography

* organism

* Time period

Alluvial soil

- Rich - potash, phosphoric acid, Lime.
- poor - Nitrogen

- Found - eastern coastal plains

- color - light grey to ash grey.

- Northern plain - 40%

↓
Indo-Gang-Brahm.

- described - on basis of age

New Alluvial

(Khadar)

old Alluvial

(Bhangar)

- crops - ① wheat ② maize

③ sugarcane

④ pulses

⑤ oil seeds.

Black Soil

- rich
 - ① Lime
 - ② Iron
 - ③ magnesia (calcium carbon)
 - ④ Alumina
 - ⑤ potash
- poor
 - ① Phosphorous
 - ② Nitrogen
 - ③ humus
- Known - Regur soil
- Black cotton soil
- clayey, deep & impermeable
- sticky
- capacity to hold moisture
- area - 15% of India.
- self ploughing
- Found - Deccan plateau
- crops -

① cotton	② MH
③ pulses	③ Guj
④ millets	④ A.P
⑤ tobacco	⑤ T.N
⑥ citrus fruits	
- black → presence of Tetranomagneteite.
- Marma → long staple cotton var.

Red & yellow soil

- rich
 - ① potash
- poor
 - ① Nitrogen
 - ② Humus
 - ③ phosphates
 - ④ lime
 - ⑤ magnesia
- Red - because ferric oxide.
- low rainfall area.
- looks yellow - occurs hydrated form.
- porous structure
- Found - E & S part of Deccan p.
 - odisha
 - chhat.

iron oxide
red.

Laterite soil

- rich - Fe & Al
- poor - N, potassium, lime, humus, potash
- develop - high temp & high rainfall
- Not suitable for cultivation
- Red laterite - suitable for tree
 - ① cashewnut.
 - TN, AP, Ke.
- Nature - Acidic soil
- found - ① KTK ② Ker. ③ TN ④ MP

Nutrient gets washed away
Leaching

Mountain soil

- rich - humus
- poor - potash, phosph., Nitro.
- Acidic in nature

Arid / Desert soil

- rich - phosphate
- poor - Nitrogen
- Highly salt content
- color - red to brown
- saline in nature
- Found - West Raj., HR, PB +
Rann of Kutch (GJ)

Peaty / marshy soil

- rich - humus
- heavy rainfall & high humidity
- good growth of vegetation
- colour - Black
- Alkaline
- large quantity of dead org. matter accumulates in area.
- Found - WB, Odisha, TN (Coastal)

Saline soils

- rich - ① sodium
② potassium
③ magnesium
- poor - ① Nitrogen
② calcium
- known - usara soils
- infertile
- occur - arid & semi arid region
- sandy to loamy ← structure range
- found - Western GJ
sunderban Area (WB)

* farmers advised to add gypsum to solve the problem of salinity in soil

current fallow

↓
uncultivated - 1yr or less

Fallow Land

↓
uncultivated - more than 5yr.

1yr - 5yr (uncultivated)
fallow other than current fallow

CROPS

* excess rainfall

- ① Rice
- ② coffee
- ③ tea
- ④ sugarcane

*

Food crop

Cereals → good Nutrition

Fine grain

- Rice
- Wheat

coarse grain

- Jowar
- Bajra (millets)
- Ragi

Pulses

- red gram
- black gram
- green gram
- pea
- soyabean

best source
of protein

Cash crop

- ① cotton
- ② Jute
- ③ Sugarcane
- ④ tobacco
- ⑤ mustard

Plantation crops

- ① Tea → ① Green tea
↓
unfermented leaves
- ② coffee
- ③ coconut
- ④ sunflower
- ⑤ mustard
- ⑥ rapeseed
- ⑦ Rubber
- ⑧ Bananas
- ⑨ cacao
- ② Black tea
↓
fermented leaves

Rainfed crop

- ① Jowar
 - ② Bajra
 - ④ millet
 - ⑤ wheat
 - ⑥ rice
- depends on rain water.
- permanent crops
- rubber
 - tea
 - coffee

Fibre crops

- ① cotton
- ② Jute
- ③ coir
- ④ hemp

Kharif	Rabi	Zaid
① grow - monsoon month	① grow - winter	① grow - summer
② June - oct/Nov.	② Nov. - march	② march - June
③ warm temp.	③ dry + cold weather	③ warm dry climate
④ wet weather	④ Ex - wheat Gram mustard Barley sunflower peas	④ Ex - watermelon pumpkin gourds veg.
⑤ Ex - paddy, maize cotton jute Jowar Bajra soyabean rice groundnut		

Dryland farming	Wetland farming
• rainfall - less • less than 75cm • Ex - ragi, moong Jowar, gram, guar, groundnut.	• rainfall - excess • Ex - rice jute sugarcane

① - WB → ① china Rice → ① staple food.	
varieties - Aus, Aman, Boro	
Kharif crop	* river valley
Wet + Warm	* coastal plain
tropical humid area	* delta

Wheat
2nd most imp. staple food.
calcium, iron, fibre
cold climate
moderate rainfall
Rabi crops
rainfall - 75cm - 100cm
N:P:K - 4:2:1
UP - largest producer in India
china - largest producer in world.

maize

- food + fodder crop
- Kharif crop
- Temp. - 21° - 27°C → wheat
- South India - Rabi crop

millet

- Ragi, jowar, Bajra
- High Nutrition value
 - Iron
 - calcium } found.
- grow less fertile area
- short duration crop (month-3)

Ragi

- Kharif crop
- mostly South India - KTK, Utt.
- Warm climate
- rainfall - 50 - 100cm

Barley

- used - food, beer, whiskey
- Rabi crop
- does not tolerate high heat
- Raj. mount. area.

cotton

- Kharif crop
- high Amount of rainfall - seed.
- Black soil
- found - Punjab, Haryana, Guj, Raj, MH, TN.
- Types - Narva (long staple cotton)
- Gujarat - largest producer India.
↓ in china

Jute

- called - golden fibre
- Hot + Humid climate
- good prod. in Alluvial soil
- cost eff.
- good strength

Sugarcane → ① Brazil

- cash crop
- belongs to - Bamboo family
- use to making - sugar
 - Jaggery (3%)
- hot & humid climate (21° - 27° C)
- Not want heavy rainfall or water.
- India - largest user & 2nd. producer

Tobacco

- 1st (came) - portuguese (1508)
- variety - ① Nicotiana Tobacum
 - ② Nicotiana Rustica
- China - largest producer & consumer.
- Gujarat - largest prod. in India

Tea

- 3/4 NE hot & humid climate.
- plantation crop
- caffeine, tannin (from 1/2)
- Assam - largest producer in India
- China - largest producer in World
- indig. to China.
- India's 1st tea plant - Assam.

Coffee

- plantation crop
- variety - ① Arabica - Superior.
 - ② Robusta
 - ③ Liberica
- max. prod. - KTK + Kerala + T.N
- Karnataka - largest producer
- Brazil - largest prod. in world.

Rubber

- plantation crop
- found - hot & humid region
- 1st rubber plantation set up
 - ↓
 - Kerala
 - ↑
 - 1st largest producer (90%)
- Thailand - largest producer in World.

Coconut

- warm & moist region
- Karnataka > TN > Kerala > A.P
- Indonesia - largest producer in World.

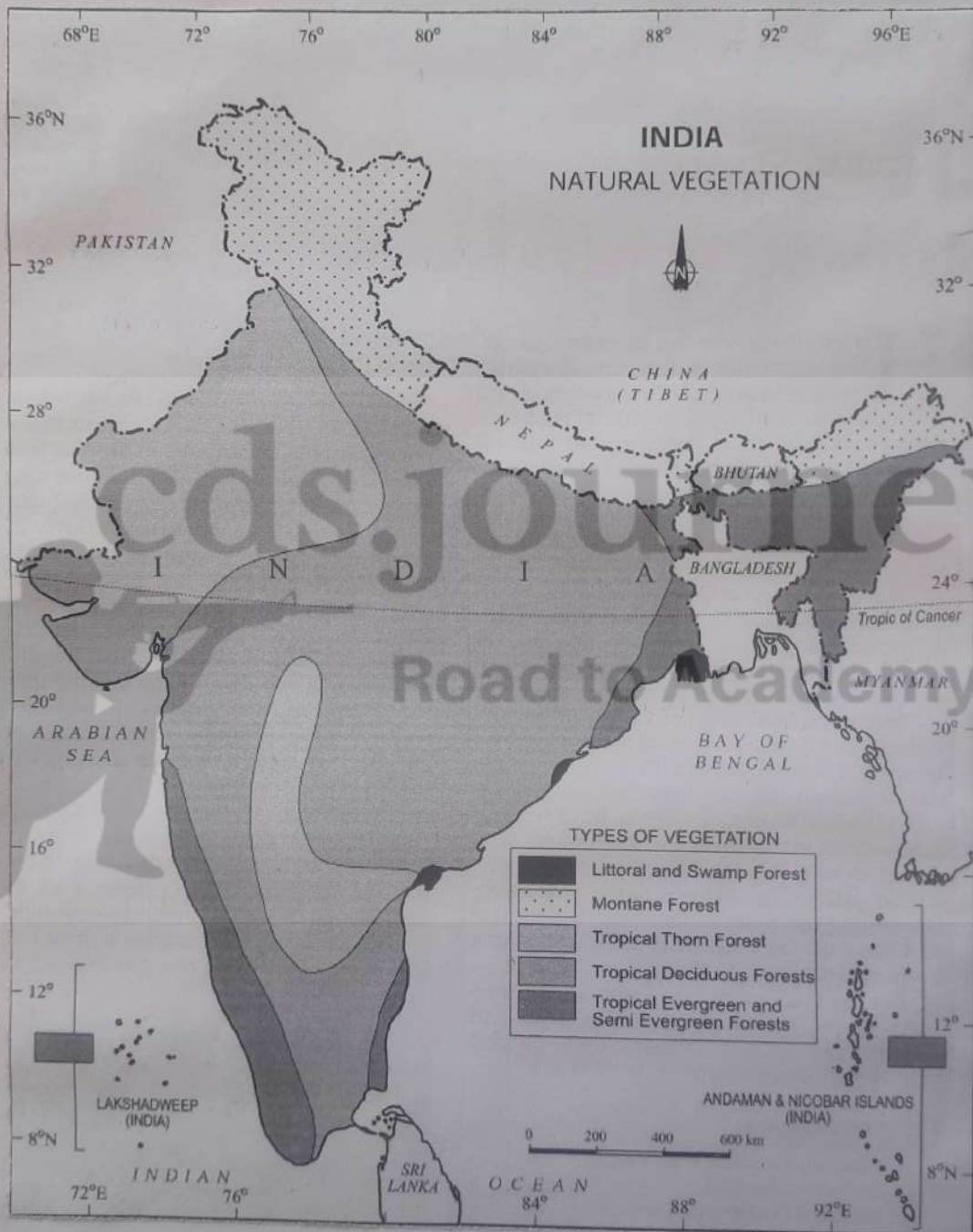
shifting cult.	countries
milpa	mexico / central America
Ladang	Indonesia / malaysia
Rocca	Brazil
caingin	Philippines
Ray	Vietnam
Tamrai	Thailand
conuco	Venezuela
chena	Srilanka
masole	Congo, Zaire (C. Africa)
Taungya	myanmar
Hay	Laos

Name	place
Jhum	Assam, NE
poonam	Kerala
podu	Andhra p., odisha
Beewar,	mip
mashan,	
penda	
Beera	

Grassland	Region
steppe	North Asia, Europe
pustaz	Hungary
Prairies	USA
pampas	Argentina
veld	South Africa
Downs	Aus.
canterbury	NZ
savannak	Africa + Aus.
Taiga	Europe + Asia
selvas	South America
Llanos	Venezuela + Colombia
campos	Brazil
manchuria	China, Mongolia

Grassland	state
mark	Kashmir
Bugyal	Uttarakhand
khajjar	H.P
ukhrul	Manipur
saramati	Nagaland
Dzukou	Uttarakhand Valley.

NATURAL VEGETATION



* vegetation types depends on precipitation.

- 200cm or more - Evergreen Rain forest
- 100-200cm - monsoon deciduous forest (moist)
- 50-100cm - dry deciduous
- 25-50cm - semi Arid
- Below 25cm - Desert (Arid)

* in India max. types of forests.

- ① Tropical dry deciduous - 39%
- ② tropical moist deciduous - 18%

Tropical Evergreen (Wet)

- maximum tree diversity
- rainfall - 250 cm +
- Temp. - 25° - 27° C
- well stratified
- trees heights - 60m or above
- do not shed leaves
- Distribution - ① western side of W.G
② A & N
③ puranchal Hills
- species : rosewood (i) mahogany
(ii) Aini (iv) ebony

Semi evergreen

- rainfall - 200cm - 250cm
- Temp - 25° - 27° C
- mixture of evergreen & moist deciduous.
- species - ① white cedar
② hollock ⑦ oak
③ kail
④ rubber
⑤ tea
⑥ bamboo

Tropical Deciduous Forest

Tropical moist Deciduous

- rain - 100 - 200cm.
- temp - Avg. 27° C
- dist - ① Foothills of Himalayas
② eastern slope of W.G
③ odisha.
- Ex - ① Teak ⑤ mahua
② sal ⑥ Amla
③ Shisham ⑦ semul
④ hurra ⑧ sandalwood
⑨ kusum.

Tropical Dry deciduous

- covers largest areas of India.
- rainfall - 70-100cm
- shed their leaves in summer.
- Transition to moist deciduous
- Found - Kashmir → Kanyakumari
 - Peninsula
 - plains of UP
 - Bihar
- Not found - Raj., W.G., Kolkata

- Ex - ① tendu ④ bel
② palas ⑤ Axlewood
③ amaltas ⑥ Khair

tussocky grass grows - height 2m

Tropical Thorn forests

- rainfall - less than 50cm
- variety of grasses + shrubs
- Found - Semi Arid Areas
 - ① south west Punjab
 - ② Haryana ③ Raj ④ Guj.
 - ⑤ m.p ⑥ U.P
- Ex: ① babool ② ber ③ Khair
④ Neem ⑤ khejri ⑥ palas

Montane forests

- rainfall - 75-125cm
- Found - Eastern Himalayas
A.P
- Ex - • chestnut
• pine forest (1500-1750m)
• sal • chil
• oak
- chinar + walnut - trees famous
Kashmir handicrafts -
- Alpine trees - found in high Altitude
(3000-4000m)
 - Fir • Junipers
 - Kali • pines
 - Rhododendrons

Littoral + Swamp forest

- also called wetland forest
- found - reservoirs of
 - ① Deccan plateau
 - ② Eastern coast deltas
 - ③ Sundarban ④ east of Guj.

Littoral

↓
shore of sea
+
lake

Swamp

↓
delta of Ganga,
Mahanadi, Cauvery,
Godavari, Krishna

mangrove forests

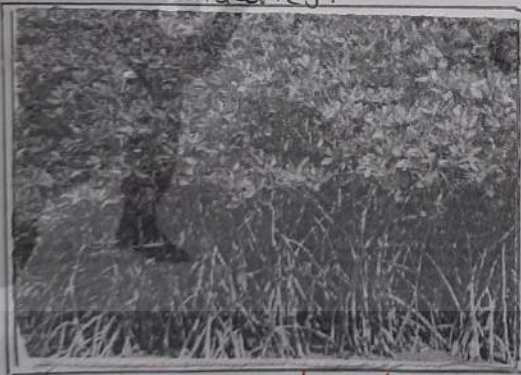
India (developed) -

- i) A & N
- ii) Sunderbans
- iii) area of Mahanadi, Godavari, Krishna deltas

7% of world's mangrove forest cover in India.

mangroves grow along coast

- marshes
- tidal creeks
- mud flats
- estuaries.



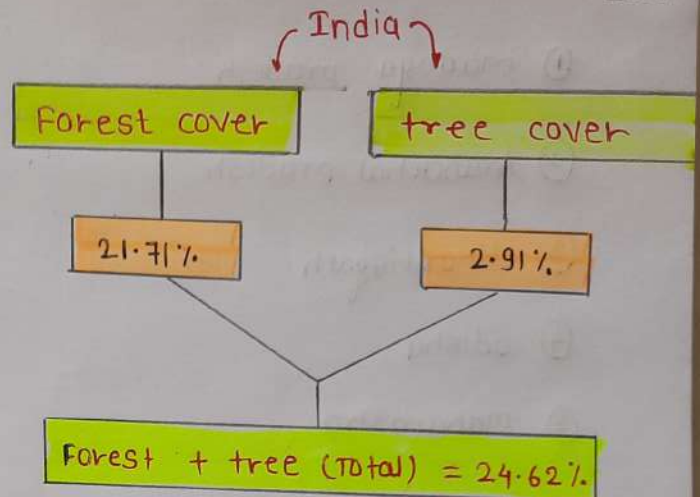
mangrove forests.

* mangrove cover

- ① Andhra Pradesh
- ② Goa
- ③ Gujarat
- ④ Karnataka

Indian state of forest Report

2021.



Top ten countries for forest Area

- ① Russia
- ② Brazil
- ③ Canada
- ④ USA
- ⑤ China
- ⑥ AUS.
- ⑦ D.R. of Congo
- ⑧ Indonesia
- ⑨ Peru
- ⑩ India

forest cover in India

class	%
↑ very dense forest	3%
moderately dense forest	9.33
↑ open forest	9.34
Non-forest	76.87

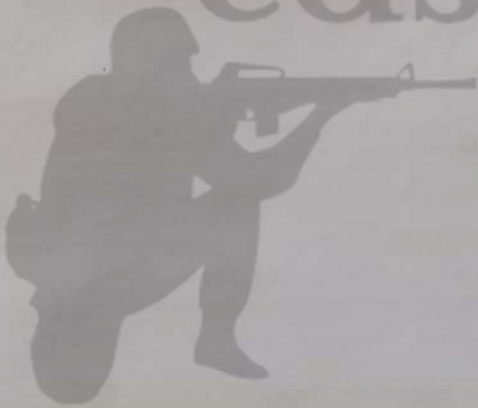
Area wise forest cover-

- ① madhya pradesh
- ② Arunachal pradesh
- ③ chhattisgarh
- ④ odisha
- ⑤ maharashtra.

% wise forest cover

- ① mizoram
- ② Arunachal pradesh
- ③ meghalaya
- ④ manipur
- ⑤ Nagaland.

cds.journey



Road to Academy...